



Human Activity Reconstruction Using Accelerometer Based Sensor Readings

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Abstract

Human activity reconstruction is widely used in the health industry, sports, rehabilitation, and virtual reality platforms. Traditional motion capture technology often relies on multiple inertial sensors including accelerometers, gyroscopes, as well as magnetometers, increasing the cost, complexity, and the power consumption. This project presents a simplified human activity reconstruction system using a sparse sensor configuration with only five 3-axis accelerometer sensors placed on the wrists, ankles, and torso which is designed as a simple wearable device. Machine learning algorithms are employed for recognizing patterns in human movement from the 15 accelerometer reading we are obtaining for a certain time period, and inverse kinematics is employed to model overall human movement. Our main aim is to represent the reconstructed model as a customized 3D avatar motion based on the body dimensions. Therefore, the proposed approach demonstrates a low cost and simple wearable alternative to conventional human activity reconstruction systems.

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Acronyms

2D Two-Dimensional

3D Three-Dimensional

4D Four-Dimensional

ACCAD Advanced Computing Center for the Arts and Design

AI Artificial Intelligence

AIEM Accelerometer-only IMU Estimation Module

AMASS Archive of Motion Capture as Surface Shapes

API Application Programming Interface

AR Augmented Reality

biRNN Bidirectional Recurrent Neural Network

CLAHE Contrast Limited Adaptive Histogram Equalization

CNN Convolutional Neural Network

CoM Center of Mass

DIP Deep Inertial Poser

DIP-IMU Deep Inertial Poser Inertial Measurement Unit Dataset

DNN Deep Neural Networks

DPVO Deep Patch Visual Odometry

ECE Expected Calibration Error

ESRGAN Enhanced Super-Resolution Generative Adversarial Network

F1 F1 Score

FID Fréchet Inception Distance

FIP Fast Inertial Poser

FPS Frames Per Second

FSR Foot Sliding Ratio

GLB GL Transmission Format Binary

glTF GL Transmission Format

GPU Graphics Processing Unit

HMR Human Mesh Recovery

HMR2 Human Mesh Recovery 2

HPC High Performance Computing

IK Inverse Kinematics

IMU Inertial Measurement Unit

KNN K-Nearest Neighbors

LAB Lightness A B Color Space

LBS Linear Blend Skinning

LPIPS Learned Perceptual Image Patch Similarity

LSTM Long Short-Term Memory

ML Machine Learning

MPJPE Mean Per Joint Position Error

MuJoCo Multi-Joint Dynamics with Contact

MVN Xsens MVN Motion Capture System

NeuralPhysCap Neural Physics-based Motion Capture

PA-MPJPE Procrustes-Aligned Mean Per Joint Position Error

PC Personal Computer

PCA Principal Component Analysis

PD Proportional Derivative

PhysCap Physics-based Motion Capture

PIFu Pixel-aligned Implicit Function

PIP Physical Inertial Poser

PP-Motion Physical-Perceptual Motion

PSD Pose Space Deformation

PVE Per Vertex Error

Real-ESRGAN Real-World Enhanced Super-Resolution Generative Adversarial Network

ResNet Residual Network
RFC Random Forest Classifier
RGB Red Green Blue
RNN Recurrent Neural Network
ROMP Regressing Monocular 3D Pose and Shape
RTX NVIDIA RTX Graphics Platform
SIP Sparse Inertial Poser
SLAM Simultaneous Localization and Mapping
SMPL Skinned Multi-Person Linear Model
SVM Support Vector Machine
TCN Temporal Convolutional Network
TIP Transformer Inertial Poser
TotalCapture TotalCapture Dataset
UV U-V Texture Coordinate
ViTDet Vision Transformer Detector
ViTPose Vision Transformer Pose Estimation
VR Virtual Reality
WHAM World-grounded Humans with Accurate Motion
WWSN Wearable Wireless Sensor Network

Chapter 1

Introduction

1.1 Background

Human activity reconstruction of a user's 3D skeletal configuration has become increasingly important for analyzing and visualizing human movement in areas such as healthcare, sports analysis, rehabilitation, virtual reality, gaming, biomechanical analysis and immersive human computer interaction including virtual and augmented reality since it enables remote monitoring and evaluation. Also, the approach of reconstructing human motion using movement data rather than using video data helps to preserve the privacy of the users of these systems. These systems must function reliably in everyday environments and minimize the intrusiveness and user instrumentation.

Traditional motion capture systems rely on high-end optical based setups with multiple cameras, markers such as Vicon, and high accuracy inertial sensors, which require expensive infrastructure, controlled environments, and complex setups to attach them to the body, making them unsuitable for portable or everyday use.

Sensor based approaches that mount devices directly on the human body can overcome the line-of-sight limitations of the vision based methods. Hence, Inertial Measurement Units (IMUs), which provide acceleration and orientation data are proposed due to their low power consumption and ease of wearability. Dense inertial configurations, typically using more than 17 IMUs placed near major joints, can directly estimate a joint's orientation. Therefore, these systems rely heavily on sensor accuracy. Even though there are commercial solutions based on this system like Xsens, dense IMU setups are intrusive, uncomfortable to wear and require extensive calibration making them undesirable for consumer level usages[1].

In contrast, recent research shows that sparse sensor configurations can effectively capture motion by using the lesser number of sensors which are placed on the key segments of the body. The reduced number of sensors leads to insufficient joint constraints, making the motion reconstruction more complex. The IMUs itself is incapable of measuring the distances directly and the early sparse IMU methods primarily based on optimization approaches which the human motion was reconstructed by matching sensor data to pre recorded motion datasets or by solving constrained convex or nonconvex optimization problems using physical and

kinematic constraints. Therefore, these methods require high computational power and present significant latency making these unsuitable for real time applications.

To address these limitations, learning based methods using Deep Neural Networks (DNNs) have been proposed and Deep Inertial Poser (DIP) was one of the earliest learning based method used to directly map sparse IMU data to full body pose using DNNs and large scale training datasets. DIP significantly improved the real time performance which was a challenge in earlier times but still had some limitations in accuracy and recreation of the complex motions. Then, the Transformer Inertial Poser (TIP) introduced transformer architectures combined with sampling based optimization to capture long term temporal dependencies in motion data improving the reconstruction quality but the computational cost was increased. Physical Inertial Poser (PIP) further incorporated physical constraints and torque modeling through dual proportional derivative (PD) controllers to model the torque and optimize the output of the transpose. But this additional optimization layer increases the computational burden and reduces the suitability for resource constrained platforms.

For real-world applications involving consumers, motion capture systems should be able to operate within mobile platforms such as AR-VR Headsets, Smart Glasses, or wearable technology that impose tight power, size, and processing resource constraints. More importantly, most of the current approaches haven't considered body shape information. This disregard causes bias to body shapes and adds to computational costs involved in pose estimation.

Then the Fast Inertial Poser (FIP) was proposed which is a real time motion capture method which can reduce the computational burden and latency while increasing the accuracy of the human activity reconstruction making it more suitable for embedded platforms with limited computational power and also it consider the body measurements of the user before wearing the device.

Also, the coupling of Machine Learning (ML) methods and biomechanical modeling has emerged as a successful strategy for the estimation of correct 3D skeletal poses using noisy and incomplete sensor data. From the point of view of IMU-based motion capture systems, Machine Learning models can be trained to learn the complex, generally nonlinear relations between the sensor data (Inertial measurements of orientation, Angular Velocity, and Acceleration) and the corresponding joint orientations of the human body. Using the prior knowledge of human motion learned by Machine Learning systems, it is possible to overcome the ambiguities inherent to the application of incomplete sensor setups and noisy sensor readings.

With joint orientation/pose parameters estimated by ML algorithms, a full-body skeletal reconstruction is often obtained via Inverse Kinematics (IK). It imposes constraints such as bone length, joint limits, and body part hierarchical order to ensure a valid skeletal reconstruction that is not only meaningful in a physical sense but anatomically feasible as well. Thus, ML is used for a problem that is undetermined yet important for pose reconstruction and temporal consistency of

the reconstruction sequence, and IK is then used to refine the final reconstruction and prepare it for visualization purposes.

The synergy of ML-based pose estimation and inverse kinematics is even more valuable for real-time computation on embedded systems. ML inference is fast once the model is trained, and IK is a light Post-processing step that maintains human body consistency with relatively inexpensive global optimization. Thus, these two combined can make skeletal data generation from sparse IMU highly efficient for minimally invasive and real-time motion capture systems used for applications like VR/AR, gaming, and human-computer interaction[2].

1.2 Problem Statement

Even though human activity reconstruction is carried out using wearable motion capture systems in areas such as healthcare, sports, virtual reality, and rehabilitation. Most existing IMU-based motion capture approaches rely on multiple sensors incorporating accelerometers and gyroscopes to estimate body orientation and joint movements with high accuracy. Commercial systems such as Xsens MVN and perception neuron typically employ 10-17 IMUs leading to increased hardware cost, complex calibration procedures and higher computational requirements for sensor fusion and drift compensation making such systems less suitable for simple and low hardware cost wearable solutions.

Gyroscope-based motion reconstruction involves sensor fusion, calibration processes, and drift compensation algorithms which increases the computational power, power consumption and the design complexity. Also, over time the drift will accumulate and this leads to inaccuracies in the orientation which affects the precision of the reconstruction over longer spans of the activity. Furthermore, when a larger number of gyroscope and accelerometer equipped sensors are used, it reduces the wearability and simplicity, making them uncomfortable for users and impractical for everyday or long-term use.

Considering all these limitations, a simple, low cost and minimally intrusive system approach is needed for human activity reconstruction. Therefore, the challenge is to utilize a sparse set of accelerometer only sensors and combine evolving engineering technologies in order to reconstruct meaningful human motion while maintaining affordability, simplicity, and user comfort.

1.3 Objectives and Scope

1.3.1 Aim

Develop a low cost and simple wearable accelerometer based human activity reconstruction system that accurately estimates the movements of the joints and reconstructs the full body motion and visualizes it through a personalized 3D

avatar while validating the reconstructed motion against synchronized reference video data.

1.3.2 Objectives

The specific objectives of the project are as follows,

1. Accurately acquire raw time synchronized motion data from the user during movement
Expected outcome – Accurate accelerometer-based sensor readings with synchronized timestamps representing the user’s physical movement to be used for further processing.
2. Design and develop a method to estimate joint postures and orientations using the collected raw accelerometer data
Expected outcome – Accurate joint posture parameters, such as joint angles and orientations, that represent the user’s physical movement for use in skeletal mapping.
3. Design a skeletal model using joint posture parameters and create a 3D avatar
Expected outcome – A realistic, 3D human avatar that reconstructs the full-body motion of the user’s physical movement.
4. Design and implement a subsystem to validate the reconstructed human activity and the avatar output through a reliable evaluation method
Expected outcome – Validation results representing the accuracy, consistency, plausibility, and reliability of the human activity reconstruction system.

1.3.3 Scope

1. Development of computational methods to estimate the posture or the movement of the human body using only accelerometer data, without reliance on gyroscopes or magnetometers. To compensate for the limited input which utilizes only 5 accelerometer based sensors, advanced predictive and calculative methods will be needed. Therefore, due to computational latency and processing constraints, real time motion reconstruction will not be feasible. Consequently, the system operates in an offline capacity; data is recorded during the session and manually processed post-hoc for activity reconstruction.
2. Integration of estimated posture or the movement into a skeletal kinematic model for a high fidelity representation of the user’s movements.

3. Creation of a personalized 3D human avatar that animates the reconstructed skeletal motion, incorporating skin generation based on individual user parameters such as limb lengths, body proportions and skin tone providing a realistic visual representation of the user’s physical movements while enhancing the user experience. However, a primary limitation of the current model is the absence of fine-grained joint articulation-specifically finger movements, subtle rotational variances, and head orientation as the reconstruction is constrained by the sparse five-sensor input.
4. Design and implementation of a dedicated validation subsystem that utilizes a synchronized video feed as the ground-truth reference. The validation process involves a side-by-side comparison of the raw video footage against the reconstructed avatar motion. This subsystem focuses on qualitative assessment and approximate accuracy rather than exhaustive quantitative precision, as the primary objective is to validate the model’s architectural integrity from a visualization perspective. The validation process may be limited by the availability, synchronization accuracy and quality of reference video data or ground-truth motion capture systems.

Chapter 2

Literature Review

2.1 Previous Work

2.1.1 Developing the Models using Sparse IMUs to Estimate the Joint Postures and the Orientations

IMU-based human activity reconstruction aims to model full-body pose while avoiding the limitations of camera-based systems. Over time, research on this topic has evolved from optimization-based approaches to deep learning models while reducing the number of sensors, computational power, cost, and improving durability. SIP, DIP, TIP, PIP, and FIP represent key milestones along this research journey.

(a) Sparse Inertial Poser (SIP)

SIP is an optimization-based method that formulates pose estimation as a global or temporal optimization problem using motion priors and kinematic constraints. Although this method achieves high accuracy with sparse IMUs, it is computationally expensive and exhibits higher latency, which limits its applicability in real-time and mobile systems [1].

(b) Deep Inertial Poser (DIP)

DIP employs bidirectional RNNs to estimate joint rotations from IMU data, incorporating learned kinematic constraints to achieve faster inference compared to SIP while maintaining good accuracy. However, this system is not suitable for real time applications because of the non causal inference and the latency of the system where it is depending on past and future frames. [3]. Figure 2.1 illustrates the overview of the DIP pipeline.

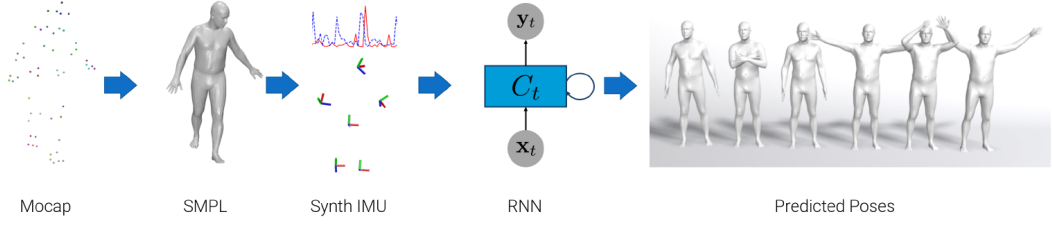


Figure 2.1: Overview of the DIP pipeline

(c) Transformer Inertial Poser (TIP)

TIP is a transformer based system which can handle complex motions accurately since it captures long range temporal dependencies without using inverse kinematics solver. But this system requires high computational power and memory resources limiting the usage in embedded and wearable applications. [1]. Figure 2.2 presents the overview of the TIP pipeline.

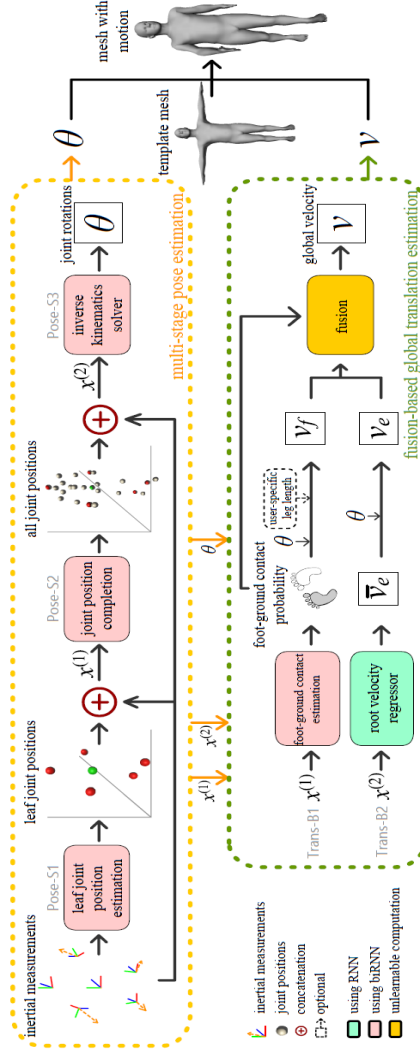


Figure 2.2: Overview of the TIP pipeline

(d) Physics-based Inertial Poser (PIP)

PIP is a physics-based model with optimization and inverse kinematics to enhance the balance constraints and ground contact consistency. This system has improved realistic human motion reconstruction and contact handling than the learning based models. But high computational power and complex system design limits the real time applicability[2].

(e) Fast Inertial Poser (FIP)

FIP is a real time human motion reconstruction model which estimates full body motion using only six sparse IMUs while considering the human body shape information. This model uses a bidirectional RNN that enables low-latency inference. Body parameters such as height and limb lengths help resolve pose ambiguities caused by sparse sensing. Experimental validation on the AMASS and DIP-IMU datasets demonstrates state-of-the-art performance, achieving over 60 FPS with less than 15 ms latency [2]. Figure 2.3 shows the overview of the FIP pipeline.

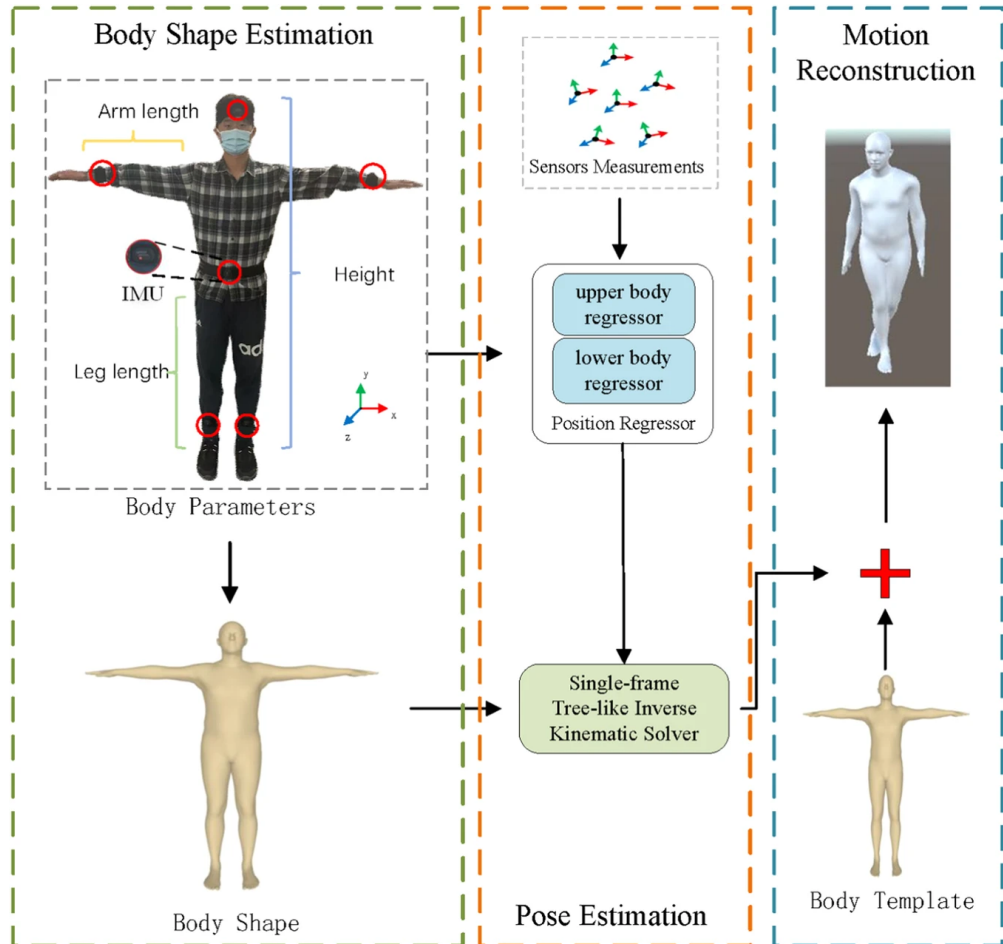


Figure 2.3: Illustration of FIP

2.1.2 Developing the Models using Accelerometers to Estimate the Joint Postures and the Orientations

1. Human activity reconstruction for full-body pose has been performed using only four sparse 3-axis accelerometers placed on the wrists and ankles. Instead of using accelerometer readings directly, which can introduce significant drift, this method used a data-driven approach leveraging a large motion capture database. A simulate-and-optimize approach was utilized to generate synthesized accelerometer readings to align with sensor data through nearest-neighbor search and graph optimization. Inverse kinematics and reduced-dimensional pose representations were used to recover plausible full-body poses in real-time [4]. Following Figure 2.4 shows the overview of the system proposed in tautges 2011.

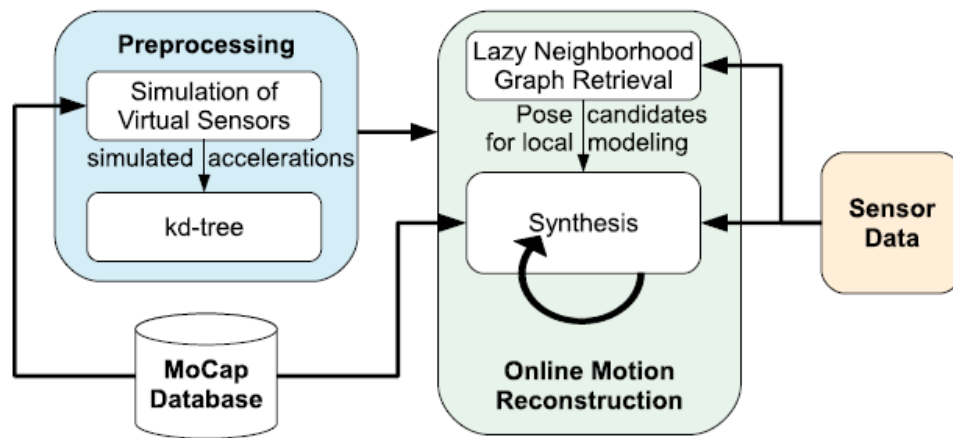


Fig. 1: *Overview of the animation system.*

Figure 2.4: Overview of the system proposed in tautges 2011

2. Human motion can also be reconstructed using wearable accelerometers in complete isolation from the requirement for optical motion capture systems. A portable, low-cost framework where several body-mounted tri-axial accelerometers were used to estimate full-body motion, especially for sports and outdoors for which optical systems are impractical. Instead of directly computing the joint angles from sensor data, their approach relies on a data-driven motion database and a motion graph constructed from pre-recorded motion capture data. On this database, virtual accelerometers were simulated, and a dynamic programming-based matching algorithm was used for finding pose sequences whose simulated accelerations best matched the measured sensor readings. While the method indeed produced naturalistic motions using only accelerometer data, it was prone to positional drift over time, showing one major limitation of accelerometer-based motion reconstruction approaches.[5] Following Figure 2.5 shows the overview of the system proposed in kelly 2010.

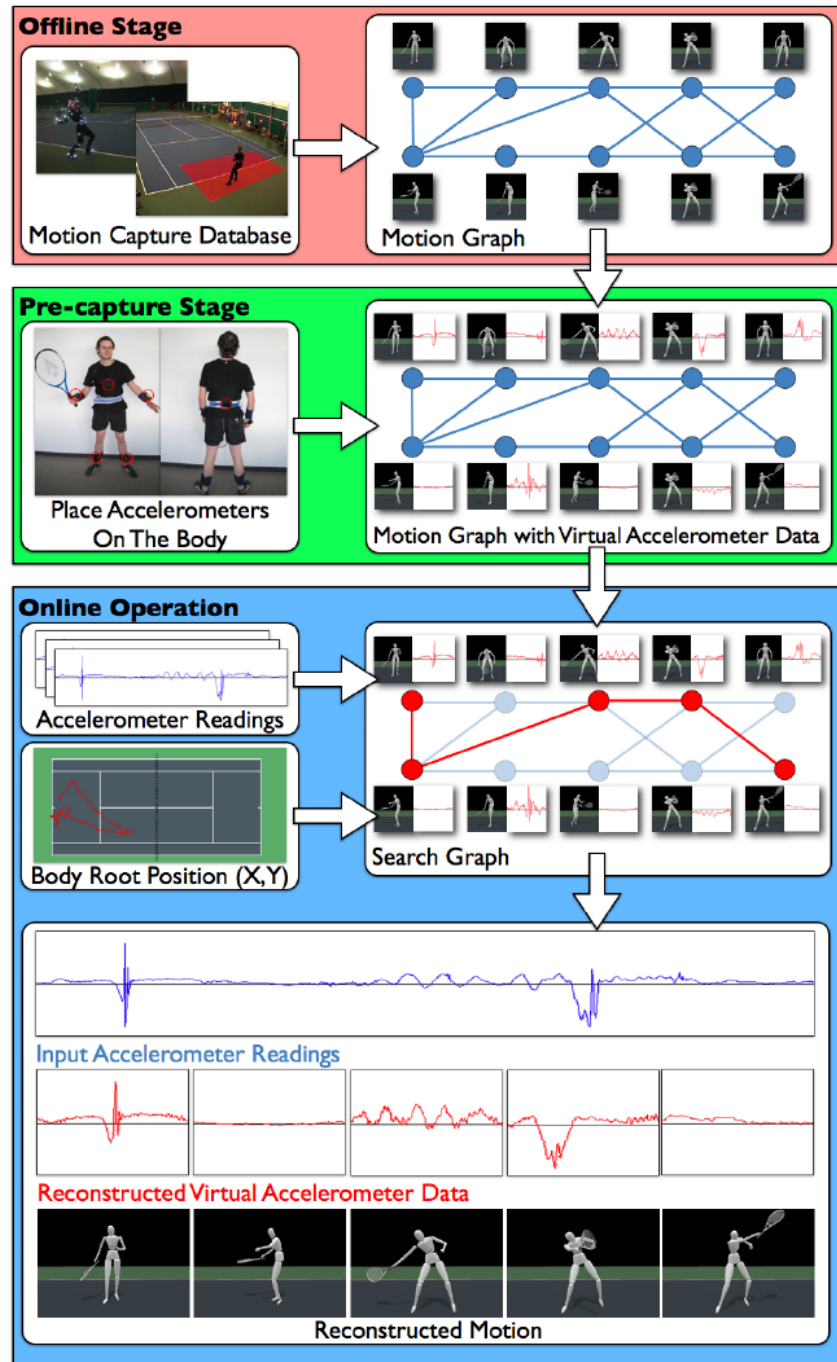


Figure 2.5: Overview of the system proposed in Kelly (2010)

2.1.3 Human Activity Detection Using WWSN and Artificial Intelligence

Beyond skeletal pose estimation, localized research has demonstrated the efficacy of a Wearable Wireless Sensor Network (WWSN) for direct activity classification using sparse accelerometer data. This study, from the University of Ruhuna, accurately differentiated six specific physical activities: cycling, pushups, running, squats, walking, and table tennis. The study also aimed to discover the optimal configurations of the sensors to lessen the burden placed on the user.

The research developed two distinct paradigms for activity classification based on the dependencies they capture:

1. Detection by Capturing Spatial Dependencies: When an activity is seen as a single snapshot of sensor readings, conventional machine learning approaches are used. The Random Forest Classifier (RFC) outperformed KNN, SVM, and Naïve Bayes, with an accuracy of over 96%.
2. Detection by Capturing Spatial and Temporal Dependencies: For activities defined as a series of instances, Convolutional Neural Networks (CNNs) were used to capture complex physical movements over time. This strategy achieved an accuracy and F1-score of more than 97%.

A key conclusion of this study was the identification of wrist sensors (Sensor IDs 1 and 4) as the most significant pair for classifying target activities. [6]. The results show that a reduced arrangement of two sensors can give reliable activity recognition with only slight reduction in performance, provided the models are trained on a balanced and sufficiently large dataset. Following Figure 2.6 shows the sensor positioning of WWSN and the three axes of accelerometer.

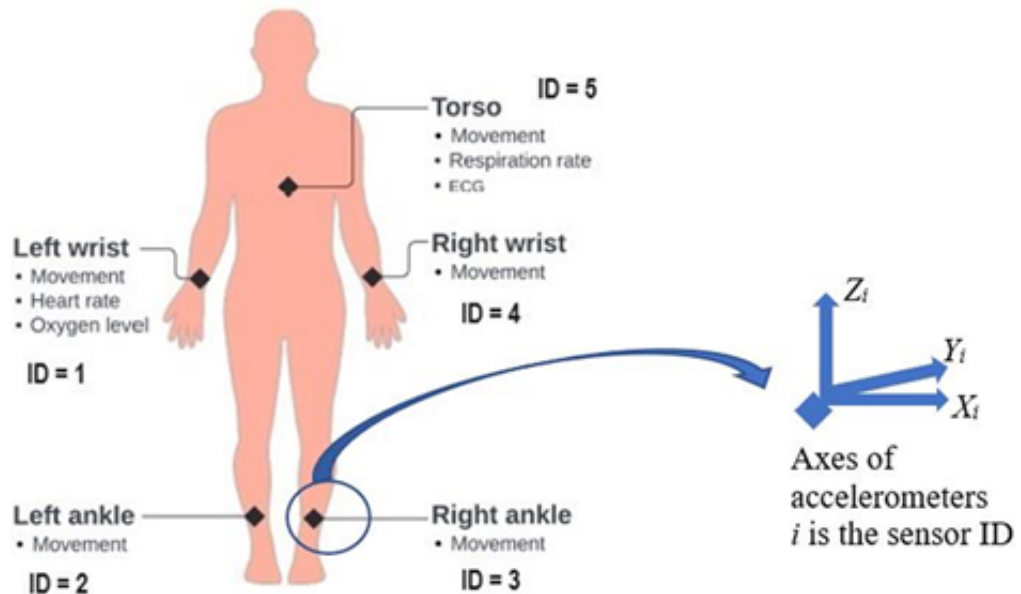


Figure 2.6: Sensor positioning of WWSN and the three axes of accelerometers.

2.1.4 3D Skeletal Reconstruction using Parametric Pose Angles

Traditional motion capture often relies on deriving absolute 3D coordinates (x, y, z) for individual joints. However, for applications requiring high anatomical fidelity, simple coordinate-based methods are insufficient as they lack surface information and rigid biomechanical constraints. Parametric skeletal reconstruction addresses these limitations by estimating pose angles of a digital skin. This approach ensures reconstructed motion adheres to physiological limits and serves as the foundation for "Digital Twin" technologies.

SMPLify (Skinned Multi-Person Linear Model)

The SMPL model is a vertex-based statistical model defining the human body surface. It is parametric, controlled by two low-dimensional vectors: Shape (β), representing coefficients derived from Principal Component Analysis (PCA), and Pose (θ), representing axis-angle rotations of joints in a kinematic tree. The skin consists of 6,890 vertices bound using Linear Blend Skinning (LBS) weights [7]. The effectiveness of this reconstruction approach is illustrated in Figure 2.7, which demonstrates the SMPLify model's performance across varying accuracy percentiles.

1. Mathematical Formulation

The body mesh is a function of shape (β) and pose (θ):

a) Shape (β)

A vector of 10 coefficients derived from Principal Component Analysis (PCA) of body scans. Changing these values alters the subject's height, weight, and muscularity without affecting their pose.

b) Pose (θ)

A vector of 24×3 parameters representing the axis-angle rotation of the 23 joints relative to their parents in the kinematic tree, plus the global orientation of the root joint (Pelvis).

2. Kinematic Tree & Skinning

The model utilizes a hierarchical kinematic tree (e.g., the shoulder moves the elbow, which moves the wrist). The "Skin" consists of **6,890 vertices** that are bound to this skeleton using **Linear Blend Skinning (LBS)** weights. This ensures that when the skeleton moves (driven by the reconstructed angles), the skin stretches and deforms realistically, providing the necessary surface data for regeneration analysis.



Figure 2.7: Evaluation of SMPLify reconstruction performance across varying accuracy percentiles

HMR (Human Mesh Recovery)

HMR utilizes a neural network to predict pose and shape coefficients from a single image in one step. This confirmed real-time skeletal reconstruction while preserving human anatomy [8]. The qualitative capabilities of this end-to-end regression are presented in Figure 2.8, demonstrating the recovery of 3D meshes from unconstrained RGB images.

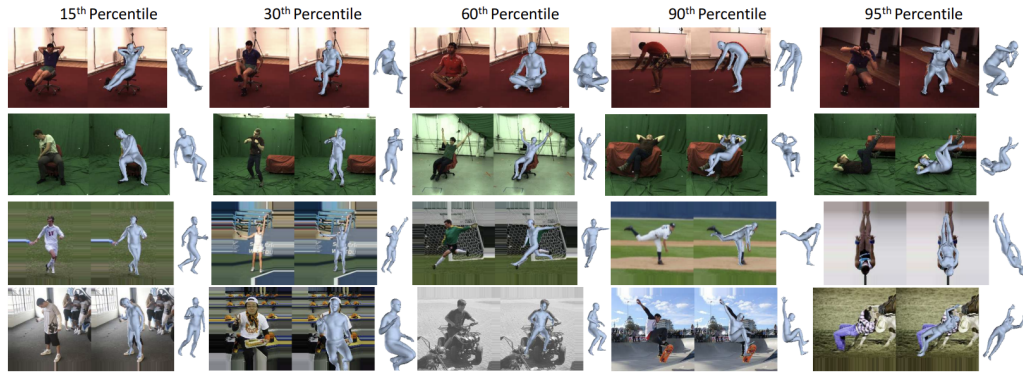


Figure 2.8: Qualitative results of Human Mesh Recovery (HMR) on unconstrained monocular RGB images

2.1.4.1 ROMP (Regressing Monocular 3D Pose and Shape)

While SMPL provides the container for the data, ROMP is the engine used to extract that data from monocular images. It is a "one-stage" regression network designed to operate efficiently on local GPU hardware[9].

1. Single-Stage Architecture

Unlike older "two-stage" methods (like SMPLify) that first detect 2D keypoints

and then separately optimize a 3D model to fit them—a slow and iterative process—ROMP utilizes a ResNet-50 backbone to perform end-to-end regression. It takes an input image and simultaneously predicts the camera parameters, body center, and SMPL parameters in a single forward pass.

2. Pixel-Level Center Map

ROMP distinguishes itself by treating the 3D body center as a pixel-level representation. It generates a "Center Map" (heatmap) to locate subjects even in crowded scenes.

3. **Parameter Map Construction:** For every detected body center, ROMP samples a feature vector that is decoded into:

- a) **3D Pose (θ):** The rotation matrices for all 24 joints.
- b) **Body Shape (β):** The coefficients describing the user's unique anatomy.

The complete one-stage pipeline, from the backbone feature extraction to the final parameter regression, is visualized in Figure 2.9, highlighting the pixel-level localization strategy.

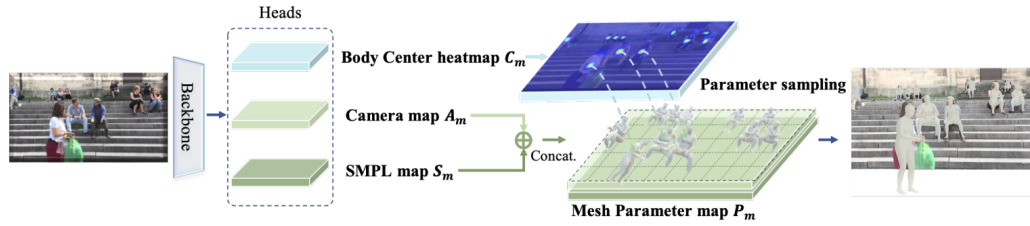


Figure 2.9: The ROMP One-Stage Regression Architecture for 3D Mesh Recovery

2.1.4.2 WHAM (World grounded Humans with Accurate Motion)

Regarding the limitations of frame-by-frame regression algorithms like ROMP, progress in the field of pose estimation using optical cameras has been achieved through temporal modeling and world-referencing. One example of such an algorithm is the algorithm called WHAM [[10]], which attempts to estimate human motion within the 3D world by referring to a common global coordinate system.

1. Temporal Sequence Modeling via Dual-Fusion

This model varies from the basic methodology of regression in stages or frames since it uses dual fusion. First, the WHAM model identifies a series of 2D keypoint features through an already trained detection method. The feature vector is then inputted to the motion encoder for learning the continuous motion of the human body. To prevent the loss of dense information from sparse keypoints that might be experienced in the first step, WHAM merges the motion information that is learned from the motion encoder with dense features obtained directly from video frames.

2. World Grounding and SLAM Integration

One of the issues associated with motion tracking using a single camera is

the problem of camera-object coupling, which involves the effect of the camera movement on the perception of the object’s movements. With regard to the WHAM framework, the angular velocity of the camera is utilized using an algorithm called the DPVO algorithm, which represents the visual SLAM algorithm. The angular velocity is input into the global trajectory decoder together with human body movement parameters, leading to the differentiation between the two in a mathematical sense.

The following diagram in 2.10, shows the architectural overview of the WHAM, demonstrating the 2D keypoint detection, dense visual features, and SLAM-derived camera angular velocity for global trajectory estimation.

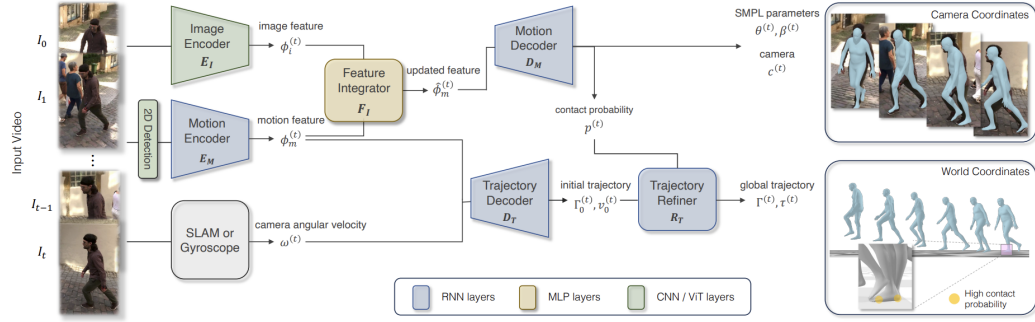


Figure 2.10: Architectural Overview of WHAM [[10]]

2.1.5 3D Skin Generation and Skeletal Binding

Human digitization has evolved from primitive skeletal models to complex, mesh-based representations containing both identity and motion. Current research in 3D skin generation and skeletal binding is dominated by three major technological pillars:

1. Parametric Body Modeling

The most influential progression in skin creation is the **Skinned Multi-Person Linear (SMPL)** model [7]. Other than the stick-figure skeleton model, SMPL is a vertex-based representation of simplified human bodies that provides a low-dimensional parameter space to describe human body shapes. It deforms a generic template with shape parameters β to correspond to distinctive physical proportion and pose parameters θ that define joint rotations. The visual separation of these parameters is depicted in Figure 2.11, illustrating how the model mathematically decouples body shape (identity) from skeletal pose (motion).

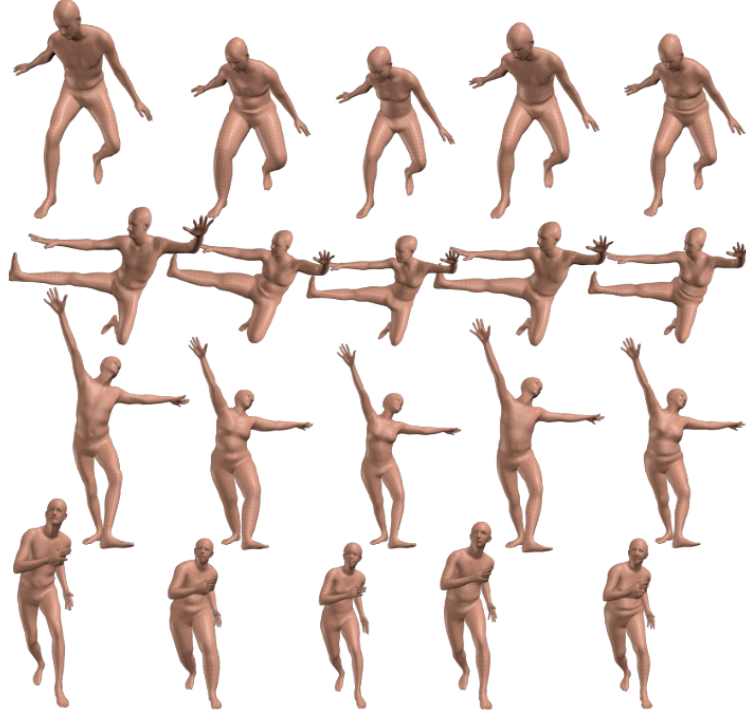


Figure 2.11: Decomposition of the SMPL model into shape (β) and pose (θ) parameters

2. Automated Mesh Recovery

The alternative to manual character creation is solved with use of the so called **Human Mesh Recovery (HMR)** methods. Research by Kanazawa et al. [8] showed that it is possible to directly regress SMPL parameters from a single RGB image using end-to-end deep learning. This regression capability is visualized in 2.12, showing the model's output on diverse input images. **Pixel-aligned Implicit Functions (PIFu)** [11] go further in this direction, enabling digitization of "clothed" humans and obtaining high-resolution texture and clothing folds that parametric models generally fail to represent. The ability of PIFu to capture these fine-grained surface details and clothing deformations is demonstrated in 2.13.



Figure 2.12: Real-time 3D Human Mesh Recovery (HMR) from a single RGB image

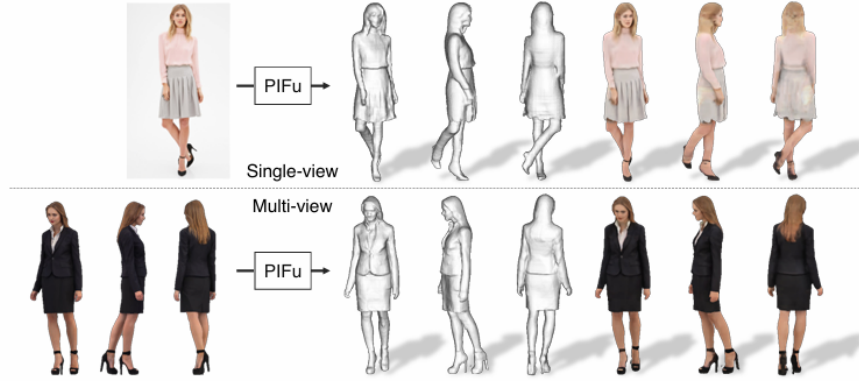


Figure 2.13: High-resolution surface reconstruction using Pixel-aligned Implicit Functions

3. Vertex-to-Bone Mathematical Binding

To ensure that the generated skin can deform with the skeleton without visual artifacts or slippage, Linear Blend Skinning (LBS) is used. LBS was developed as skeletal subspace deformation. While basic LBS can suffer from volume loss, the foundational work in Pose Space Deformation (PSD) [12] created the mathematical means to connect the weighted influence of mesh vertices to the motions of the bones. This balances the surface deformation to the active bone movements while reducing joint distortion artifacts. The improvement in volume preservation offered by this technique compared to standard skinning is illustrated in Figure 2.14.



Figure 2.14: Comparison of joint volume loss in standard Linear Blend Skinning (LBS) vs. the corrected Pose Space Deformation

2.1.6 Validation for Accelerometer-Based Activity Reconstruction

A primary objective of this research is to validate the proposed activity reconstruction model through a comprehensive evaluation of its accuracy, consistency, plausibility, and reliability. To ensure rigorous validation, the study utilizes a hierarchical framework consisting of four distinct levels, each targeting a specific dimension of reconstruction quality. This multi-level approach is particularly essential for accelerometer-based systems, as inherent challenges such as sensor drift and the limited observational capability of inertial measurement.

i. **Geometric Accuracy Assessment:**

Geometric validation forms the foundation of motion reconstruction evaluation, primarily relying on Euclidean distance metrics to quantify positional deviations from ground truth data obtained through optical motion capture systems[13].

- a) **MPJPE (Mean Per Joint Position Error):** The average distance (in millimeters) between predicted and true joint coordinates.
- b) **PA-MPJPE (Procrustes-Aligned MPJPE):** This rigid alignment removes errors related to global rotation and scale, focusing purely on the "pose" structure.
- c) **PVE (Per Vertex Error):** For full avatars (meshes), this measures the error of every vertex on the body surface, capturing body shape (e.g., fat/muscle) accuracy better than just joints.

ii. **Physical Plausibility Verification:**

Given that accelerometers fundamentally measure force-related quantities governed by Newtonian mechanics, validation must extend beyond geometric correspondence to encompass physical realism[14].

- a) **Foot Sliding (FSR):** A critical metric for generative models. It measures the distance feet travel while they should be planted on the ground. High foot sliding is a primary indicator of poor reconstruction quality.
- b) **PhysCap & NeuralPhysCap:** These methods validate motion by running it through a physics simulator (like MuJoCo). If the simulator cannot replicate the motion without applying impossible external forces (e.g., invisible strings pulling the puppet), the reconstruction is flagged as "implausible".
- c) **Center of Mass (CoM) Stability:** This verifies if the avatar is balanced or if it would physically fall over given the reconstructed pose.

iii. **Perceptual Quality Evaluation:**

Recent scholarship acknowledges that mathematical accuracy does not guarantee perceptually natural motion, addressing the potential for technically correct but visually uncanny results[13].

- a) **LPIPS (Learned Perceptual Image Patch Similarity):** A deep-learning metric used to compare rendered frames of the avatar against real video. It correlates better with human visual perception than pixel-level errors.
- b) **FID (Fréchet Inception Distance):** Measures the distance between the distribution of generated motions and real human motions. A lower FID means the motion "looks" more natural and human-like.
- c) **PP-Motion:** A newly proposed metric (2025) that combines physical constraints with human perceptual scores to give a single "Fidelity" rating.

iv. **Reliability and Uncertainty Quantification:**

The most sophisticated validation tier addresses model confidence calibration and epistemic uncertainty. This reliability assessment is particularly pertinent for accelerometer-based systems operating in unconstrained environments where activity diversity may exceed training data coverage[15].

- a) **Expected Calibration Error (ECE)** measures the alignment between predicted confidence levels and actual accuracy rates, with elevated ECE values signaling unreliable confidence estimates.
- b) **Epistemic uncertainty quantification** identifies knowledge gaps in the model’s training distribution, proving essential for detecting when the system encounters novel activities beyond its learned repertoire.

2.2 Gaps in Literature

2.2.1 Developing the Models using Accelerometers to Estimate the Joint Postures and the Orientations

There are several gaps remaining in accelerometer-only based motion human activity reconstruction existing research. First, dependence on large fixed motion databases like AMASS limits the generalization ability and the quality of reconstruction of out-of-distribution motions as they fail to reconstruct complex and novel movements that were not present in the training data which leads to stiff motions. Second, optimization and retrieval methods being used in earlier approaches for motion reconstruction require high computational costs limiting the applicability in lightweight systems. Additionally, existing accelerometer based methods face ambiguities in rotational pose estimation and require very strong assumptions in the case of missing information when it comes to yaw rotations. The fine grained motion information, global translation estimation and adaption capability for different human morphologies and complex motions are also less represented in the existing research. These facts point towards more sophisticated methods being designed for more effective and efficient research in human activity reconstruction using accelerometers[4].

2.2.2 Parametric Reconstruction and Anatomical Binding

While notable progress has been achieved separately in the areas of 3D surface modeling and inertial pose estimation, effectively combining these two technologies remains a major challenge—particularly when operating with sparse sensor setups. The following sections outline the specific deficiencies in current research that this study aims to address.

Metric Scale and Manual Measurement Ambiguity

Current regression-based approaches such as ROMP[9] and HMR[8] are capable of reliably estimating 3D body shape and proportions; however, their outputs are typically expressed in a unit-less coordinate space. Most existing works focus

on visual pose appearance rather than absolute metric correctness. This limitation is a major unresolved challenge for clinical applications, such as skin regeneration, where real-world measurements are essential. Moreover, existing sparse inertial systems such as the Fast Inertial Poser[2] still rely heavily on manual measurements. Users are required to input pre-measured limb lengths and biological parameters to properly calibrate the skeletal model. Currently there is no fully autonomous pipeline that bridges automated data derived from single-perspective visual inputs with the real-time requirements of sparse inertial sensing

Biometric Mismatch and Kinematic Drift

A recurring problem in motion reconstruction is the Body Size Mismatch caused by the use of "standard" digital templates. Research shows that even slight differences between a digital skeleton's segment lengths (e.g., the humerus) and a user's actual anatomy can cause the Inverse Kinematics (IK) solver to compute inaccurate joint angles[16]. This often leads to kinematic drift and visually unrealistic artifacts, like feet sinking through the floor. Current studies do not offer a reliable method for automatically resizing a digital skeleton to match a user's specific limb lengths using just a single-photo calibration before inertial tracking.

Volume Preservation and Physiological Binding Gaps

The final challenge lies in maintaining the visual realism of a 3D surface during extreme motions. Conventional techniques for skeletal binding, like Linear Blend Skinning (LBS), often struggle with "volume loss"[12]. This results in joints—particularly elbows and knees—appearing to collapse or form a pinched, "candy-wrapper" effect when bent deeply. While generic corrective models exist, they fail to consider the unique muscle and soft-tissue volumes of an individual. Currently, there is no integrated approach that leverages a user's initial photograph to guide the 3D model in realistically stretching and folding the skin according to that person's specific anatomy.

Table 2.1: Summary of Identified Gaps in Literature

Gap Category	Current Limitation	Impact on Reconstruction
Metric Ambiguity	Models are unit-less and require manual height/limb entry.	Lack of clinical accuracy in skin regeneration.
Biometric Mismatch	Reliance on generic "average" skeletons.	Calculated joint angles are mathematically incorrect.
Binding Artifacts	Standard LBS causes joint collapse and volume loss.	Unrealistic visual representation during movement.

Chapter 3

Methodology

3.1 Research Design

This research follows a structured approach to human activity reconstruction. First, joint postures and orientations are estimated using sensor data and computational models. In parallel, a 3D skeletal model is reconstructed, and a virtual avatar is generated to represent human motion. These two outputs are integrated to form a complete motion representation. Finally, the proposed system is validated using experimental data, and its performance is evaluated based on accuracy, consistency, and reliability metrics. Following Figure 3.1 illustrates the proposed methodology.

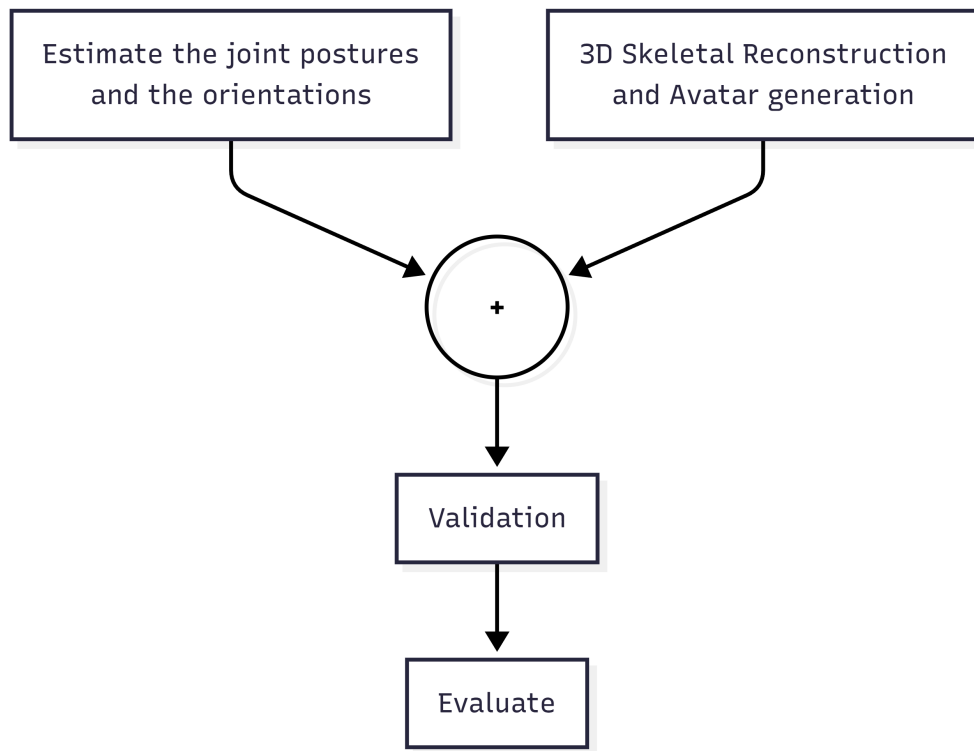


Figure 3.1: Block diagram of the proposed methodology

3.1.0.1 Estimate the joint postures and the orientations

The flow chart in the following figure 3.2 presents the full architecture of the human motion reconstruction algorithm based on accelerometers only. Full architecture consists of three stages: Sensor Rotation Estimation Model which is the newly added sensor rotation prediction layer to approximate the unknown sensor rotations resulting from the lack of gyroscope information, FIP (Fast Inertial Poser) which is used to obtain the joint rotations, and the final human motion output.

The unprocessed data from six accelerometers is initially fed into the Sensor Rotation Estimation Model, in which the physics-based and deep-learning-based models are tested for their ability to predict the missing sensor rotations. The optimal model is chosen depending on the minimum value of the rotation prediction error. At last, the predicted sensor rotations along with the unprocessed data from accelerometers are used as input to the Fast Inertial Poser (FIP) model.

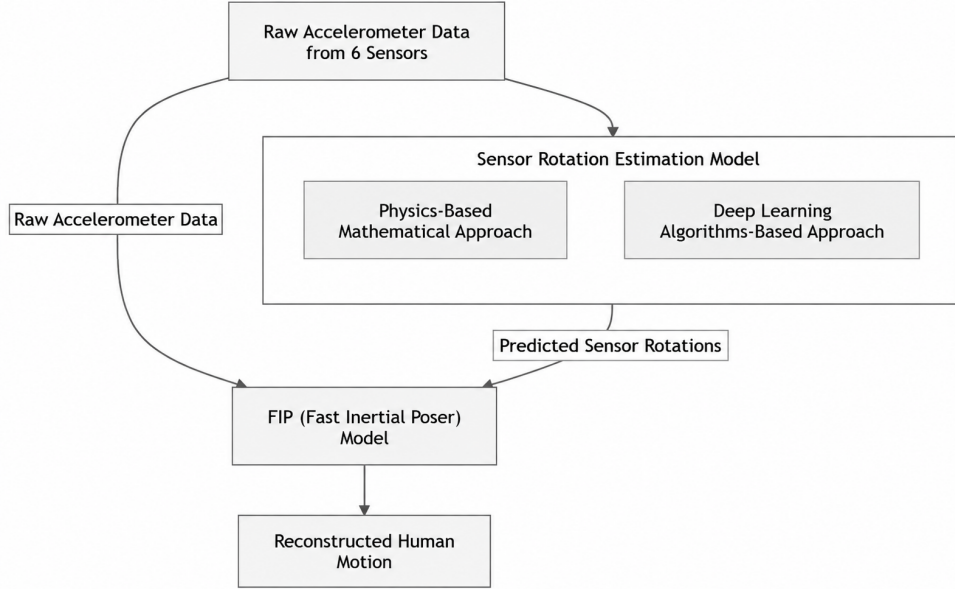


Figure 3.2: Pipeline of the proposed model

1. Deep Learning Based Rotation Prediction Models

Two deep learning-based approaches were explored to estimate the joint rotations.

For both the approaches AMASS was preferred over DIP-IMU and TotalCapture datasets as there was an increase in diversity regarding the range of motion that was recorded by human beings. In addition, sensor data generation became convenient through the use of the SMPL body model. Moreover, sensor data generation became very convenient since the positions of the sensors were free to choose without dealing with IMU calibration problems.

i. Hybrid Inverse Kinematics and Deep Learning Sensor Rotation Estimation Model

The suggested technique employs inverse kinematics and deep learning methods to predict the rotations of body joints using sparse accelerometer signals. The training of temporal deep learning models was achieved

using artificial acceleration measurements created based on the AMASS dataset to learn the dynamics of the body movements. The entire framework consists of data generation, preprocessing, creating sliding window sequences, predicting rotations using inverse kinematics and deep learning methods, and finally estimating human poses using the SMPL body model. The following flowchart in Figure 3.3 illustrates the overall pipeline of the proposed frame-level orientation estimation approach using accelerometer data.

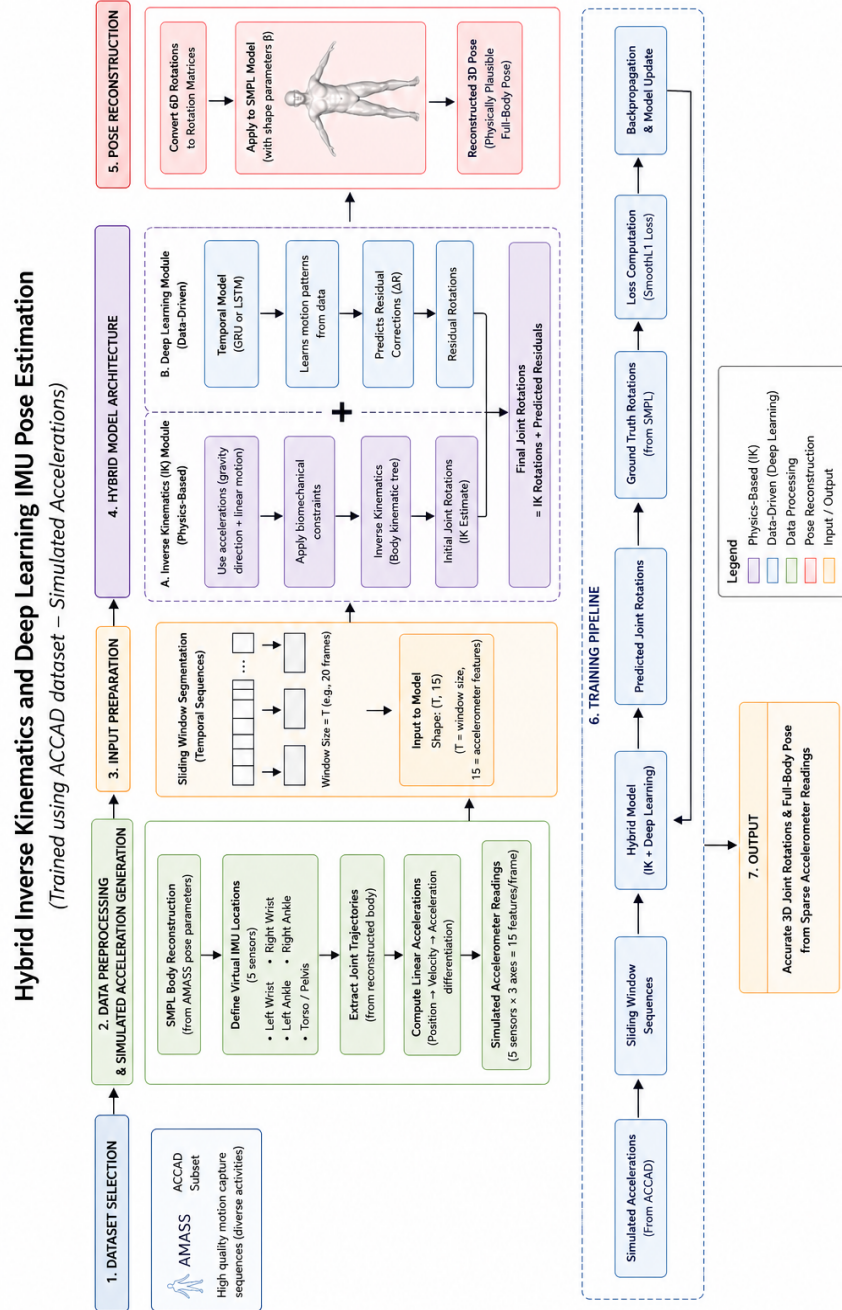


Figure 3.3: Overall Pipeline of the proposed Hybrid Inverse Kinematics and Deep Learning Sensor Rotation Estimation

To begin with, a sub-set from the large dataset named ACCAD was used to train the model, and since it was performed in our own PCs, only a small dataset was needed for preprocessing and training.

The sliding window technique was employed to incorporate the time dependency of human actions through grouping of consecutive measurements from the accelerometer into short sequences, rather than analyzing single frames in isolation. This allowed the deep learning model to learn motion continuity and temporal relationships between movements, improving the accuracy and stability of the predicted body rotations.

In this case, a light weight and physics based approach was applied to estimating the pitch and roll angles using gravitational alignment of the acceleration signal, thus resulting in a highly efficient yet very easy to train model. It performed well for joints like the pelvis where gravitational signals proved to be very helpful. However, the major disadvantage with the model was that it failed to perform well due to lack of ability on the part of the accelerometers to detect the yaw angle, thus making it ineffective especially in twist and turn actions, particularly in the ankles and wrists.

ii. **Sequence-Level Orientation Estimation using Attention-Based Deep Learning**

While the physics-based model showed the feasibility of using accelerometer measurements alone for estimating device orientation, there were some issues to be solved in this regard. First, there was a lack of estimation capability for yaw rotation; second, a narrow temporal context was present at a local level; third, estimation is done sensor-wise independently. The goal of the Accelerometer-only IMU Estimation Module (AIEM), which uses deep learning models, is to eliminate the drawbacks of the model using gravity vector alignment algorithm. The AIEM makes inferences about full three-dimensional rotations, including yaw rotation, using only the sequences learned from accelerometer measurements. The following flowchart in Figure 3.4 illustrates the overall pipeline of the proposed sequence-level orientation estimation from accelerometer data.

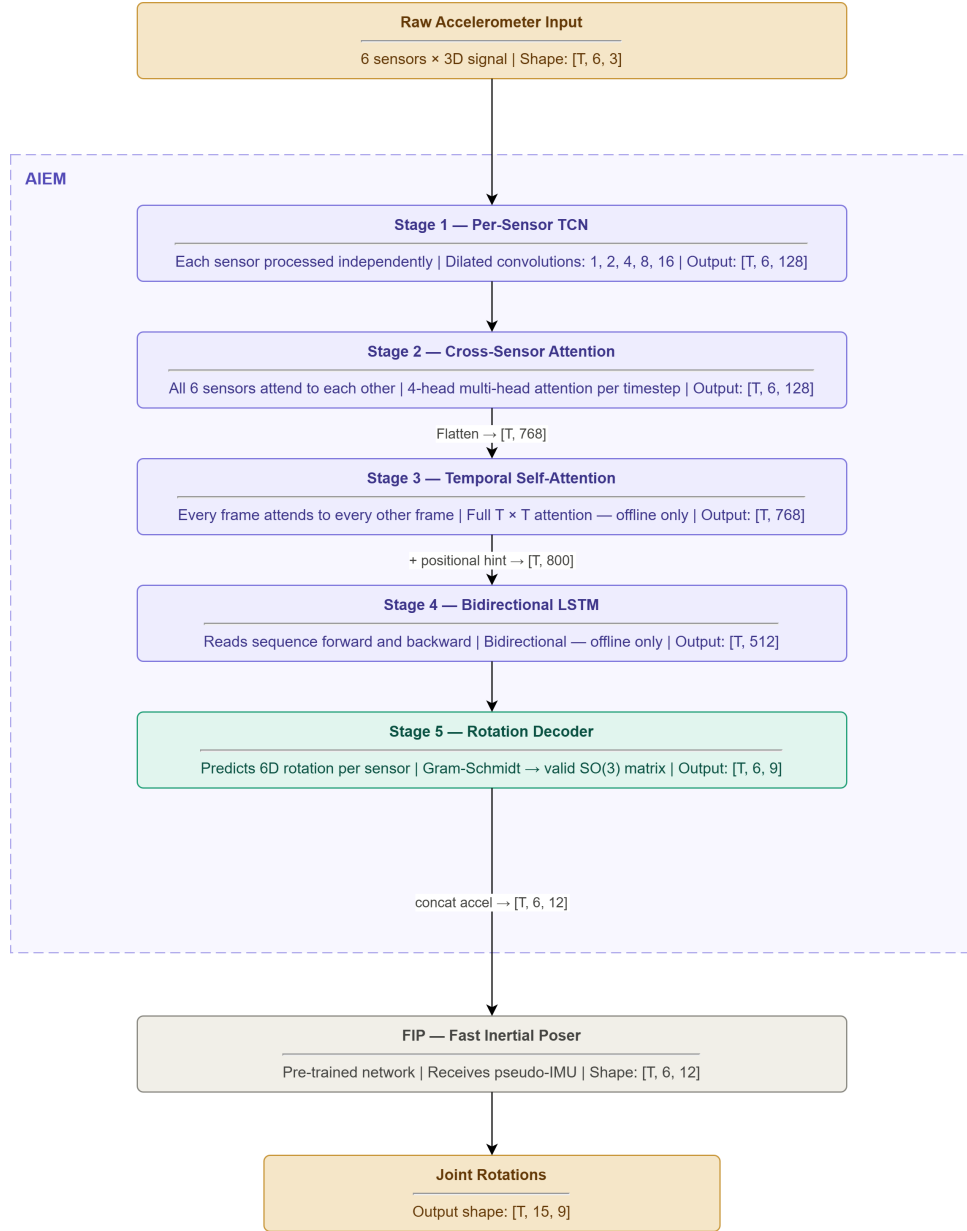


Figure 3.4: Overall Pipeline of Sequence-Level Orientation Estimation using Attention-Based Deep Learning

Following are the main stages of the AIEM architecture of the above mentioned approach.

Temporal Convolutional Network (TCN)

In the TCN, each of the six accelerometer sensor data is processed independently. Dilated convolutions depending on time are applied. In other words, it means that simultaneously with the analysis of time intervals located close to each other, remote time intervals are also taken into account. As a result, motion characteristics are identified based on the three-dimensional acceleration data, and each output is transformed into a 128-dimensional vector.

Cross Sensor Attention

Once the representations from individual sensors have been generated, there is a cross-sensor attention layer wherein all six sensors can communicate between each other. During each time point, the sensor observes the states of the other five sensors and updates its state according to the same. This is due to the real-world relationship between different body parts. For instance, the state of the pelvis affects how the ankle could possibly be oriented.

Temporal Self-Attention

The six sensors' data is flattened into a tensor and then self-attention is applied to the entire temporal dimension. The idea here is that all frames can be considered relative to each other so the pose classification has global context over the entire sequence. This process is especially helpful in resolving ambiguous poses which become clear when considering the entire trajectory of motion.

Bi-directional LSTM

The Bi-directional LSTM model processes inputs in two directions; from left to right (forward direction) and from right to left (backward direction). Both of these models process information to give an output, which makes sure that the output considers information about its past and future states to provide a better orientation estimation of the whole sequence.

Rotation Decoder

The last module uses this high-dimensional vector to generate rotation matrices from six parameters, which requires converting the 512-dimensional vector into rotation matrix space through the Gram-Schmidt orthogonalization method.

2. Geometric equations for rotation estimation

The current investigation has established a mathematical model of studying human arm motion using a two link kinematic model. As opposed to the deep learning algorithm, which relies on heavy training and large datasets, the model utilizes the basic geometric and trigonometric principles. Reachable workspace is calculated by using the theory of spherical shell while Rodrigues' formula is used for calculating the joint positions from the pre-defined angles, in the case of forward kinematics. The motion reconstruction could be carried out in real time by breaking down the gravity measurement from the accelerometer into roll and pitch angles. The following flowchart in the Figure 3.5 shows the architecture of the approach mentioned above.

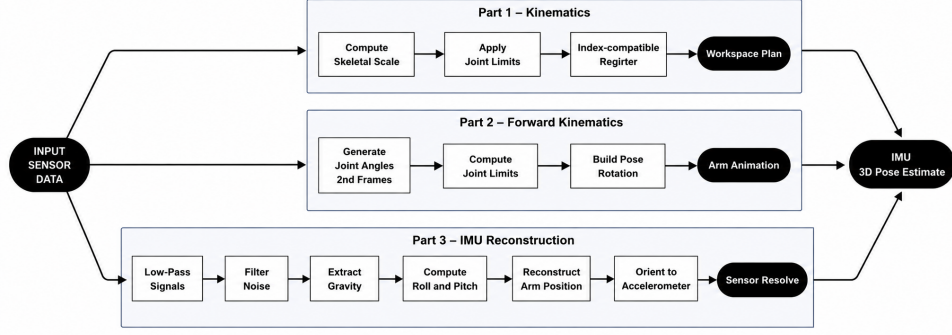


Figure 3.5: Pipeline of Geometric Equations for Rotation Estimation Model

Part 1 - Reachable Workspace

This part describes the workspace of the arm in terms of a spherical shell, which helps compute the range of reachability of the arm using mathematical constraints to determine the maximum and minimum reachable distances and visualize them as spheres.

Part 2 - Forward Kinematics

This part demonstrates realistic arm movement using joint angles. By applying spherical coordinate systems and the Rodrigues Rotation Formula, the exact 3D coordinates of the joints are computed and animated over 200 frames.

Part 3 - IMU Sensors Data Reconstruction

This part reconstructs motion using accelerometer data obtained from the wrist IMU sensor. The sensor data is filtered using moving average and gravity decomposition algorithms to estimate roll and pitch angles and compute the wrist joint position in 3D space without using machine learning algorithms.

There are some of the advantages offered by this mathematical model. First, no training data is needed to use this model. The amount of computational power needed is also very little. The process remains comprehensible at all times during computations. However, this approach does have certain limitations. It is not possible to compute yaw angles with the help of accelerometers. Fast dynamics rule out any possibility of using gravitational measures in estimating angles. In addition, it uses approximations to get the joint location. More sensors will be needed to construct the entire state space.

3.1.1 Validation

For the purpose of carrying out an effective evaluation of the correctness of the orientation determination models using the accelerometers, there has been set up a visual verification method. The objective in this case is to determine reference joint theta values using the visual data as a benchmark for comparison against the sensor-based estimates.

1. The Baseline Model (ROMP)

The initial experiments using visual reconstruction were conducted using ROMP, which is a single stage regressor for obtaining multiple poses of a person in three-dimensional space. ROMP has been used in this experiment, where it was run on videos, and the obtained pose from the SMPL model was saved in a format consisting of 72 axis angles for each frame, along with the global orientation angle for the entire body. Since there is no temporal modeling done while extracting the theta sequence, it can be presumed that there would be temporal jitter in the estimations of the pose.

2. Upgraded Architecture (WHAM)

To overcome the drawbacks that exist in the independent frame regression framework, the validation back-end was modified to include the WHAM framework. The WHAM framework has two critical advantages compared to the initial framework:

a) Temporal Sequence Modeling

By implementing temporal sequencing using WHAM, the ability to predict motions across frames can be utilized; therefore, ensuring kinematic consistency and eliminating the jitters that come from a one-step regressor.

b) World Grounding via SLAM

Because of the addition of DPVO, which assists in tracking the camera’s position, WHAM can differentiate between camera movement and human motion.

3. Robust Vision Preprocessing

The performance of WHAM is limited by the precision of the 2D keypoint detections that the underlying detector model (ViTPose) produces. Given that 2D skeleton detection models are highly susceptible to issues like motion blur, low lighting conditions, and camera shake, it was decided to incorporate a GPU-based preprocessing stage into the model.

a) Ego-Motion Compensation and Subject Isolation

An ego-motion correction step has been added to avoid motion blur due to camera motion. It uses the optical flow algorithm known as Lucas-Kanade method[[17]] for identifying features in the background and estimating the affine transformation matrix, which mathematically cancels out any motion done by the camera. After stabilization, a maximum pooled motion saliency mask is generated to differentiate the human from the background.

b) Adaptive Illumination and Local Contrast Recovery

In the case where data collection is carried out under uncontrolled conditions, the effect of non-uniform lighting will be present, resulting in low-quality gradient cues for Vision Transformers. To counteract the problem, the pipeline adopts the step of transforming individual images into LAB color space. In this process, CLAHE[[18]] is applied to only the L-channel. Contrast improvement is done on a localized basis, thus making the object more visible in dark regions without having the negative effects of histogram equalization.

c) **Edge-Preserving, Denoising and Landmark Sharpening**

Noise reduction is the final stage of preprocessing that does not distort the borders of the figure. In particular, Guided Image Filter [[19]] is performed first to eliminate high frequency noise from the camera sensor but retain strong boundaries of the body shape. Second, straight away after filtering, unsharp masking is used to highlight visible features of the body contours – namely, fingers, wrists, and ankles. This is especially vital since such high-frequency areas are needed as keypoint nodes in the ViTPose[[20]] detector’s kinematic tree.

3.1.2 Avatar Generation and Animation

3.1.2.1 Digital Twin Initialization

Overview and Objective

One of the key aims of the Human Activity Reconstruction project is the visualization of the kinematically predicted data from the sensors using a personalized digital twin. To accomplish this, a specially-designed software pipeline was developed that generates a fully textured, accurate 3D avatar from a single uncalibrated, front-on 2D photograph.

This component is used to define the static visual asset (body mesh and texture) which will be dynamically animated later by the kinematics engine. The generation pipeline is automated and runs through six steps, all of which use cutting edge computer vision and deep learning frameworks.

Pipeline Architecture and Methodology

1. Stage 1: Image Enhancement and Pre-processing

The output 3D texture quality, as well as the accuracy of the biometric estimates made by the neural network, is largely determined by the resolution of the original photograph. The pipeline automatically determines the size of the image. When a short edge of the input image is identified as being less than a certain threshold (512 pixels), the system sends the image to a Real-ESRGAN (Enhanced Super-Resolution Generative Adversarial Network) [21]. A $4\times$ up-scaling algorithm designed to retain fine edges and facial details, ensuring that the resulting pixel grid for the feature extraction process has a high density. The raw baseline input to this enhancement process is the typical front-facing photograph shown in Figure 3.6.



Figure 3.6: The standard, front-facing 2D photograph utilized as the raw input for avatar generation.

2. Stage 2: Person Detection and Body Shape Estimation

The system utilizes the HMR2 (Human Mesh Recovery) architecture [22] in the “4D-Humans” project to mathematically model the body shape of the subject.

a) Detection

A ViTDet (Vision Transformer Detector) [23] is used to perform a scan of the image and find the human image in the image and mark it out in a precise bounding box.

b) Estimation

The processed image is fed to the HMR2 transformer model that predicts the parameters of the cropped image into the SMPL body model (Skinned Multi-Person Linear) [7]. The pipeline specifically leverages 10 β (beta) parameters, every one of which is connected to distinct body features, for example, height, shoulder width, or limb length.

3. Stage 3: SMPL Mesh Deformation

The master SMPL body template is deformed to fit the proportions of the user through the extracted beta parameters. In neutral form, the SMPL template has 6,890 vertices and 13,776 triangles. The beta parameters modify the template mesh in a particular direction, for instance, making the body taller, wider or changing the limbs’ lengths. Each beta value is applied to the template, and the resulting mesh is unique to the body shape of the person in the picture. The neutral state of this master template (before this subject-specific deformation) is shown in Figure 3.7.

4. Stage 4: Person Segmentation

The subject needs to be separated from its physical context before photographic colors can be mapped to the 3D model. The pipeline includes a deep learning background removal module to produce a very accurate binary alpha mask. This step helps eliminate any non-human pixels from the 2D image, which could otherwise be incorrectly baked onto the avatar’s clothes or skin.

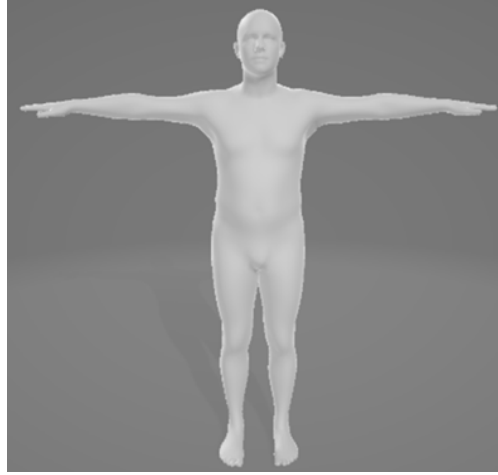


Figure 3.7: The neutral master SMPL 3D template prior to biometric deformation.

5. Stage 5: UV Unwrapping and Texture Baking

This stage creates a high-resolution 2048×2048 texture atlas by carefully composing the user's specific skin and clothing onto the 3D mesh, which is carried out in four steps:

a) **Parameterization**

The xatlas library unwraps the 3D surface into a non-overlapping 2D UV coordinate space.

b) **Color Projection**

RGB values are extracted from the photograph and directly projected on top of the same area of the UV atlas.

c) **Obscured Surface Processing**

Front-facing surface elements are displayed with all of the original colours of the picture as rendered in the photo; side-facing and hidden surface elements seamlessly fade to a "Body Color," which is an average derived from the image.

d) **Refinement**

Final post processing filters remove boundary gaps and surface noise, leaving a clean, photo realistic texture.

6. Stage 6: GLB Assembly and Export

The 3D mesh is exported as a personalized and textured mesh in 3D format (GLB). GLB is a binary version of the glTF (GL Transmission Format) standard, a popular 3D file format that packages the geometry (3D mesh), UV coordinates, texture atlas, and material properties into a single, self-contained file. The format allows it to be played on a standard 3D viewer, a game engine (e.g. Unity, Unreal Engine), or a web-based rendering library (e.g. Three.js) and therefore can be integrated into the planned web part of the system. The finalized, personalized digital twin resulting from this complete generation pipeline is displayed in Figure 3.8, demonstrating the successful application of the deformed mesh and baked texture atlas.



Figure 3.8: The finalized, personalized 3D digital twin generated from the 2D input photograph.

3.1.2.2 Kinematic Motion Application

1. Data Integration and Preprocessing

The kinematic motion application employs an interactive visualization engine built with React, Three.js, and React Three Fiber that runs on the web browser. The initial step involved in the animation is loading asynchronously the raw data files of the motion, which include the shape parameters (`betas.json`), joint angles (`thetas.csv`), and world translations (`trans.csv`). During the process of parsing the 72-column joint angles data, a moving average filter of a 2-window size is employed. It is important to smooth out the data at this point to avoid any floating point precision errors due to serialization.

2. 3D Pose Reconstruction and Quaternion Conversion

The time series obtained are used for building the animation using the SMPL skeletal structure. The 72-dimensional vector comprises the orientations (axis-angle representation) of 24 joints in the skeleton. The axis-angle representation is automatically converted to Quaternions while performing the 3D rendering process iteratively (the calculations take place in the `useFrame` function). The reason behind this conversion is related to avoiding problems that might occur because of the Euler angle rotation representation. Namely, Euler angle suffers from gimbal lock problems and is highly sensitive to the sequence of rotation angles. Finally, the absolute translation position is scaled to ensure that its difference with respect to the $t = 0$ time is removed. As a result, the digital twin always starts at the origin of the 3D world independently of the person's position relative to the camera.

3. Temporal Playback Engine

To make sure that the timing of the generated digital motion matches the actual physical motion in real life, the playback engine is kept separate from the rendering process, which is hardware-dependent. While the 3D scene render is executed continuously using the `requestAnimationFrame` loop, the time status of the avatar is handled by the interval timer, which runs at 30 frames per

second. Even though the JavaScript loop in the browser cannot provide millisecond accuracy, the architecture makes it possible to prevent the acceleration of the movement due to a fast client refresh rate.

A visual representation of the implemented web interface is provided in Figure 3.9.



Figure 3.9: The web-based motion viewer interface for visualizing the animated digital twin.

3.2 Data Collection

3.2.1 Accelerometer Based Sensor Readings Collection

1. Five 3-axis accelerometer sensors are placed on key body segments: both wrists, both ankles, and the torso.
2. Each sensor measures linear acceleration along three orthogonal axes (x, y, z), including both gravitational and motion-induced acceleration.
3. Prior to acquisition, sensor calibration corrects offset and scale errors. Sensors are securely attached to minimize relative motion.
4. Sensor axes are aligned with anatomical reference frames, and signals are sampled at a fixed frequency as continuous time-series data.
5. Predefined activity sets ensure consistency across data samples.

3.2.2 Biometric and Kinematic Data Integration

This module defines the inputs required to build the Digital Twin:

1. **Visual Calibration Data:** A single high-resolution monocular RGB image is captured with the subject standing in a neutral A-pose to minimize self-occlusion. The pipeline makes use of a dynamic resolution check to deliver

high fidelity feature landmarks to the neural network. The shortest edge of the image must be at least 512 pixels in length or will be processed through a Real-ESRGAN network to enhance it by $4\times$ with AI.

2. **Manual Biometric Metadata:** The only manual metadata required by the pipeline is the subject's gender. This classification is a primary routing control to use the correct baseline SMPL template (Male or Female). It sets the proper biological topology, meaning that the base skeletal structure, shoulder-to-hip ratios and physiological fat distributions are inherent and correct prior to any further shape deformation by the neural network.
3. **Kinematic Interface:** The system acts as the receiving unit for the dynamic movements pipe line. It requires a 72 dimensional orientation vector (consisting of 24 distinct skeletal joints in axis angle form) as its input per frame. Currently, it has been configured in a manner such that it takes in the true kinematic data that is acquired through the optical pipeline of WHAM. This is done in order to assess the performance of the digital twin and figure out the exact data structure for the sensor pipeline component.

Chapter 4

Results

4.1 Estimate the joint postures and the orientations

However, both geometric equation based approach for estimating rotations and Hybrid Inverse Kinematics with Deep Learning method for estimating rotations had some issues including inability of detecting the yaw angle, inability to understand the motion dependency on time, and generation of unrealistically physical motions. Therefore, among all other techniques used in estimating the rotations, the Sequence Level Orientation Estimation technique with attention-based deep learning showed the best accuracy. Consequently, development and testing of the whole model was done using this technique.

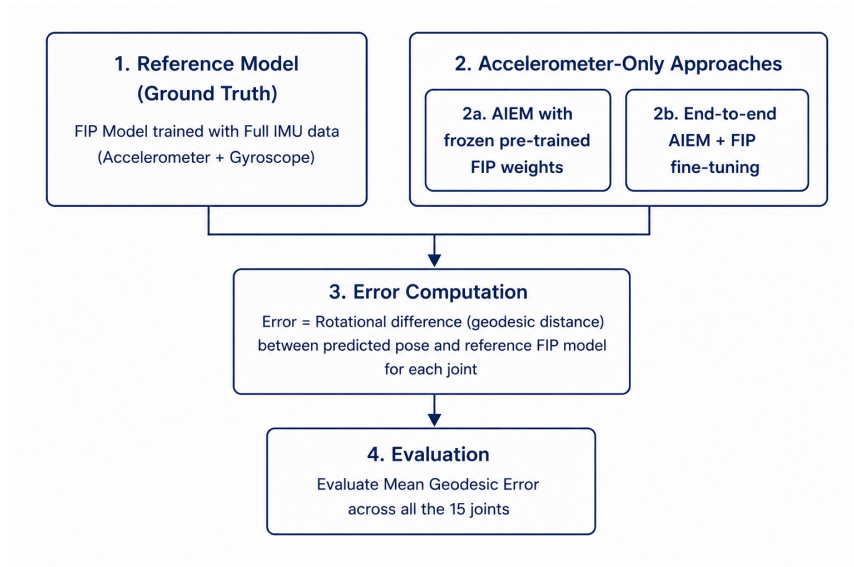


Figure 4.1: Validation Pipeline Using the DIP IMU Data set

Figure 4.1 above illustrates the validation pipeline that was adopted to validate the motion reconstruction techniques introduced using only the accelerometer sensors relying on the DIP-IMU dataset. In this context, the FIP model trained with IMU data, including both accelerometer and gyroscope signals, was selected as the reference technique. Two approaches relying on accelerometer-based signals were

evaluated. In the first case, AIEM signals were provided together with pre-trained weights for the FIP model, while, in the latter one, both models were fine-tuned in an end-to-end manner.

The geodesic error for each joint using the two approaches that use only accelerometers has been plotted on the below graph in Figure 4.2. As indicated by the graph above, the darker regions signify higher rotational error. As can be deduced from the graph above, the end-to-end AIEM + FIP fine-tuning approach had much lower errors compared to the approach with frozen pre-trained FIP weights in nearly all joints. Notably, the approach with frozen pre-trained FIP weights exhibited very high rotational error rates in some joints, particularly those on the shoulder, elbow, hip, and knee.

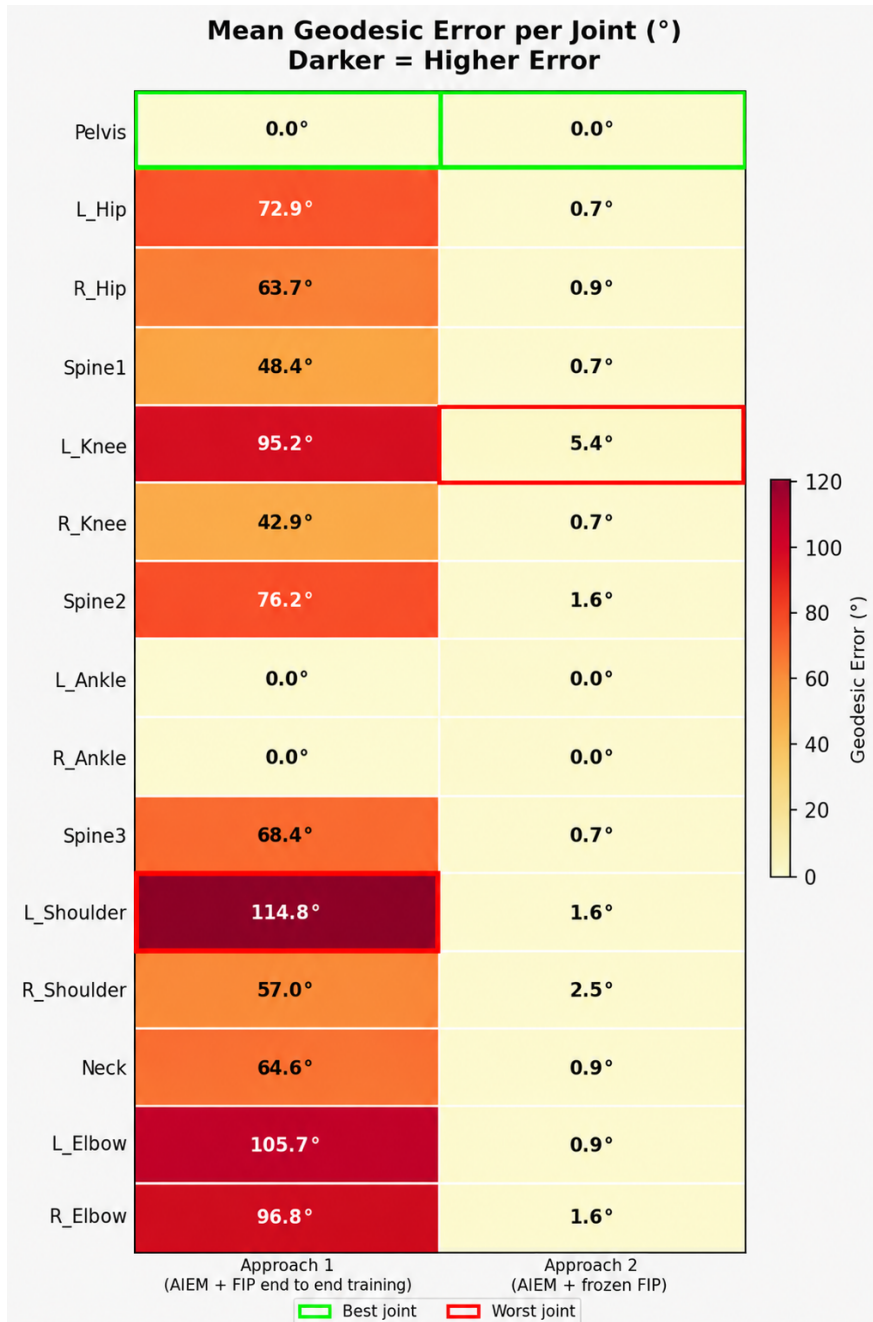


Figure 4.2: Graph of Mean Geodesic Error Per Joint for both the Approaches.

Therefore, further development and training were continued using Approach 2, as it achieved higher accuracy and more reliable performance compared to the Approach 1.

As follows from the chart in the Figure 4.3 below, there was the reduction in SIP(Sparse Inertial Poser (SIP) metric used to evaluate motion reconstruction error). error changes during the various experiments of the proposed approach of sequence-level orientation estimation through attention-based deep learning. Hence, it is possible to see the reduction of SIP error values from the first run, when the model provided the worst results, up to the optimal model. As we can see from this chart, the SIP error of the first experiment was very high and reached 102° . With regard to the following models, there was the decrease of the SIP error up to 30° and 28° correspondingly. The latest run provided us with the mean SIP of 25.9° calculated on 1,321 motion sequences.

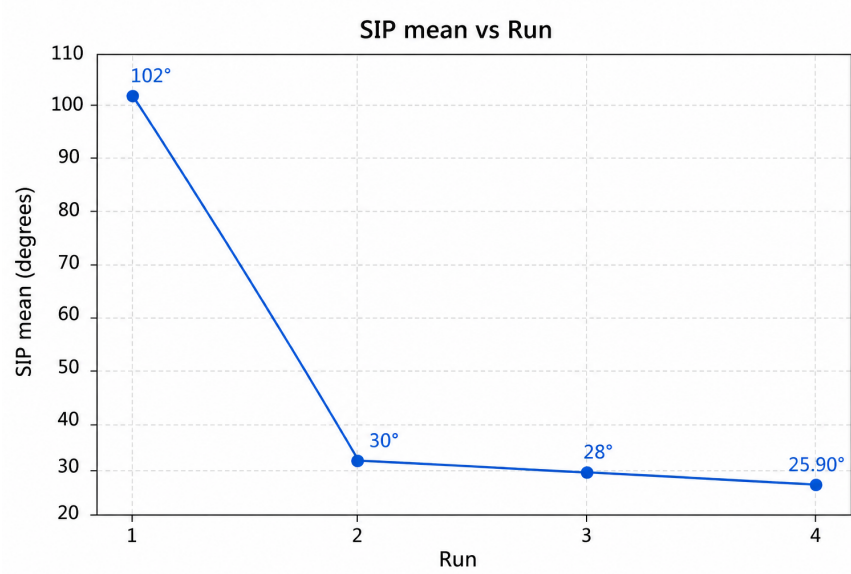


Figure 4.3: Graph of SIP mean vs the Number of Runs

Nevertheless, compared to the FIP model with IMU data of all axes, the model that only uses accelerometer readings demonstrated a larger SIP error (11.5° higher) than the latter one. The reason for this was the absence of the yaw angle and its angular speed.

Table 4.1: Typical SIP values for different motion categories.

Motion Category	Typical SIP	Why
Balance recovery, slow walks	7–15°	Predictable acceleration patterns, low dynamics
Normal walking, gentle gestures	15–20°	Periodic, learnable motion cycles
Jogging, side-steps, turns	20–30°	Higher dynamics, some directional ambiguity
Running, jumps, lifts	30–40°	Strong non-linear accelerations
Circular walks	40–60°	Continuous rotation – impossible without gyro
Hard throws, high kicks	60–72°	Maximum velocity, orientation changes fastest

The above table gives us more insights into the effects that different motions have on the SIP values. The slow motions, such as balancing and walking, will generate lower SIP values because their motion is periodic in nature and they have almost no dynamics. Walking and gestures will result in smaller SIP errors since their motion patterns are also periodic and known. However, motions that are dynamic in nature like jogging, running, jumping, powerful throw, and spinning will generate higher SIP values since their acceleration is highly non-linear and they also contain a lot of rotations.

4.1.1 Validation - Visual Reconstruction Performance

The shift from a frame-based regressor method (ROMP)[[9]] to a sequential modeling technique (WHAM)[[10]] yielded significant improvements in efficiency, stability, and consistency. In order to evaluate the improvements achieved by the model, a performance assessment experiment was conducted with a standardized sprinting sequence, which consisted of 371 frames.

4.1.1.1 Preprocessing Enhancement and Subject Isolation

Prior to evaluating the deep learning libraries, the qualitative analysis of the custom vision pre-processing pipeline was performed using the benchmark video sequence 'sprint' with 371 images. It was done to make sure that the custom vision pipeline will restore localized contrast and kinematic landmarks without adding any background noise.

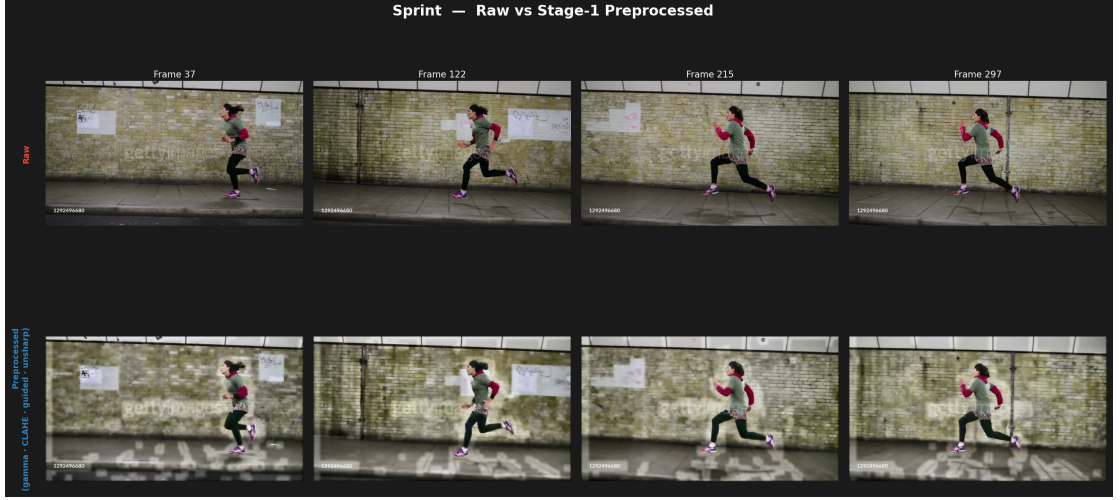


Figure 4.4: Qualitative comparison of raw video frames versus Stage-1 preprocessed frames

The first issue with the original footage is that it is an underexposed shot with low contrast, as can be seen in Figure 4.4. In the preprocessed image sequences, it is visible that there is an obvious increase in the size of the area in relation to the object. This proves that the ego-motion stabilization using the Lucas-Kanade approach worked out well, which allowed applying the mask of saliency to extract the figure of the running person, which, in its turn, facilitated applying the correction only to the needed areas.

4.1.1.2 Computational Efficiency

Processing of the 371 frame video sequence was conducted using the standard ROMP architecture that operates locally with a throughput of 6.38 Frames Per Second (FPS). The above mentioned throughput rate is inadequate for conducting sensor validation at high frequency intervals. In contrast, processing the exact inference using the WHAM architecture that employs the RTX 5090 HPC backend server was completed within 0.84 seconds with a throughput of 441.14 FPS.

4.1.1.3 Temporal Stability and Jitter Reduction

The major disadvantage associated with the monocular baseline approach was that it involved a high occurrence of temporal jitter. It was measured based on the absolute difference between the successive frame orientations ($\Delta\theta$), for all 72 orientation channels (24 joints in axis-angle form).

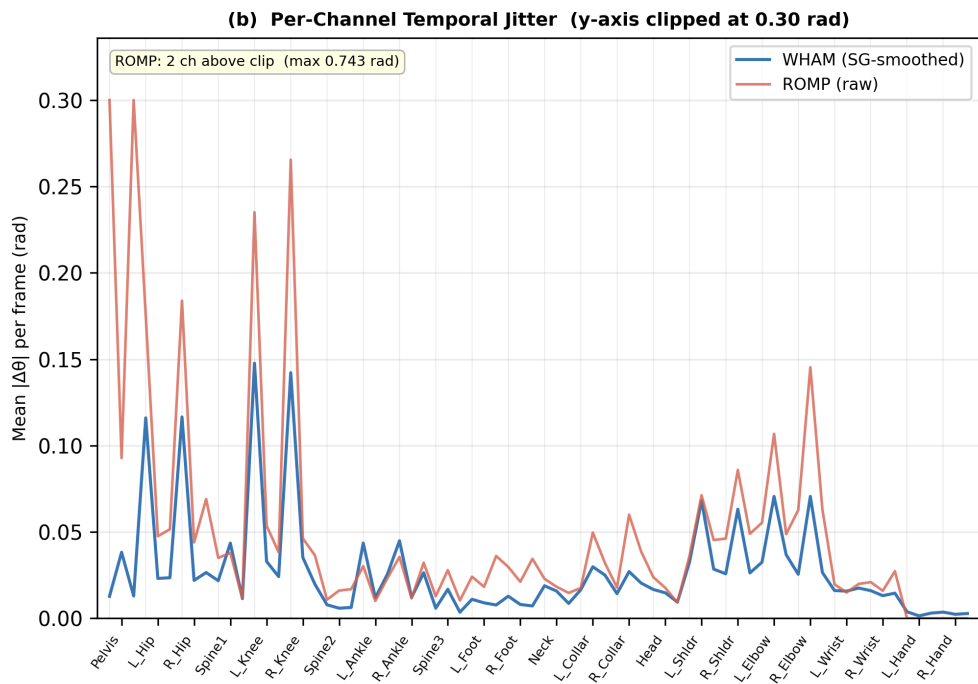
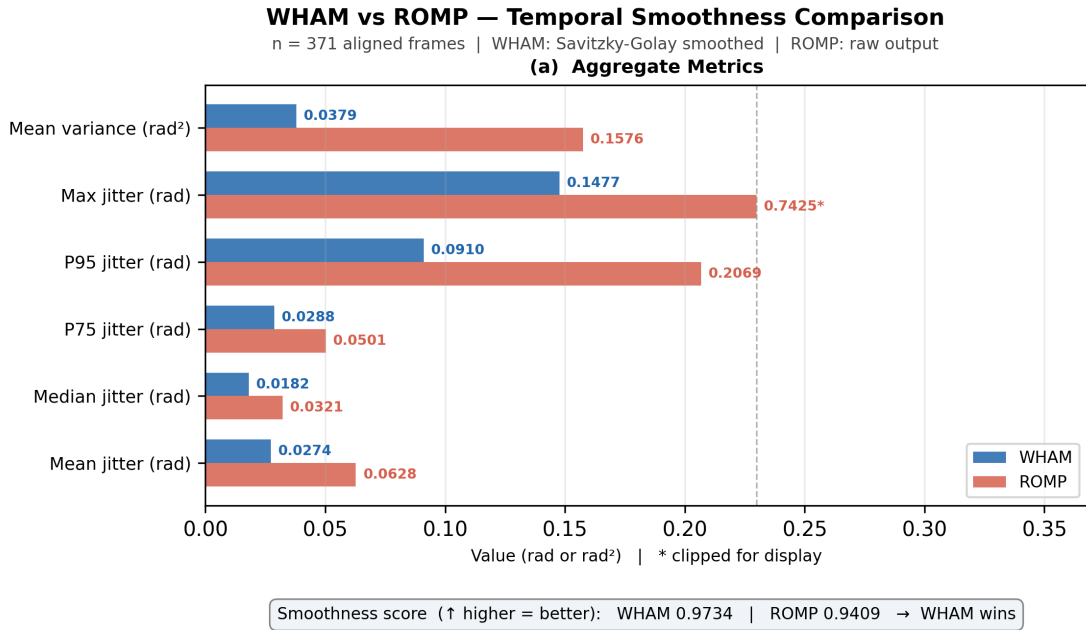


Figure 4.5: Figure 4.1: Temporal Smoothness Comparison. (a) Aggregate metrics. (b) Per-channel temporal jitter

As demonstrated in Figure 4.5a above, WHAM outperformed ROMP in all stability metrics. Specifically, the mean temporal jitter metric dropped by more than 56 %/ from 0.0628 radians per frame for ROMP to 0.0274 radians per frame for WHAM. Additionally, WHAM drastically reduced the impact of outlier values. The largest individual jitter value recorded in ROMP was 0.7425 radians, which resulted in too much teleportation of limbs. However, in WHAM, the highest

individual jitter value was just 0.1477 radians.

The assessment of the jitter based on per-channel analysis assists in identifying the exact regions where the model breaks down when performing highly dynamic motion. In the SMPL hierarchy model, the Pelvis serves as the root joint, and any error in rotation at the root gets compounded throughout the hierarchy. Looking at Figure 4.5b above, one can notice that the ROMP hierarchy suffers catastrophic failures in relation to the Pelvis, with a jitter that is too high for visual perception (0.743 radians). Also, high articulating joints during the sprint stance (left knee and right knee) suffer critical failures leading to limb teleportation.

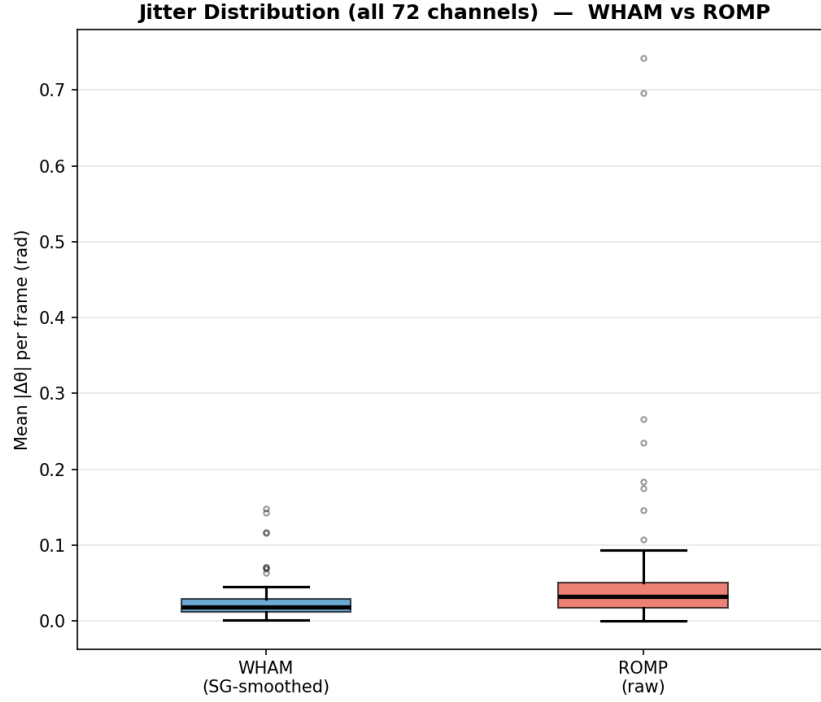


Figure 4.6: Jitter Distribution Boxplot illustrating the suppression of extreme orientational outliers

The error distribution for WHAM, as can be seen in Figure 4.6, displays that the error distribution for WHAM has a narrow interquartile range without any outliers; however, since ROMP is independent of any frame of reference, its error distribution is unpredictable.

4.1.1.4 Preprocessing Enhancement and Subject Isolation

According to joint analysis of jitter, there exists one particular area of failure of the traditional method that is corrected by WHAM.

Per-Joint Temporal Jitter — WHAM vs. ROMP Sprint sequence (n = 371 frames) | sorted by ROMP jitter

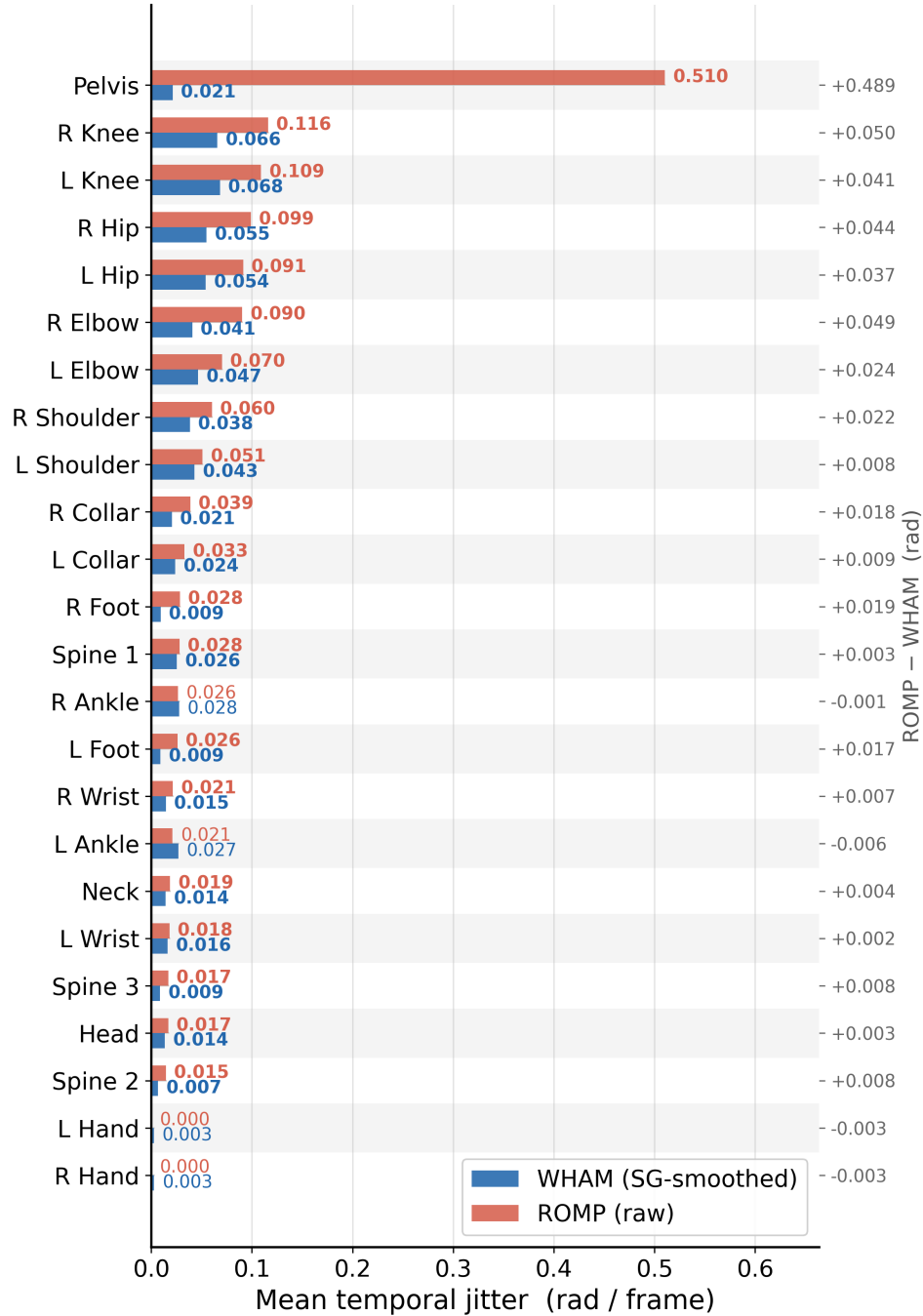


Figure 4.7: Per-Joint Temporal Jitter in ROMP vs WHAM

SMPL model kinematics uses the joint named Pelvis as the root joint. For the baseline approach, a mistake made in the root joint leads to exponential amplification of jitter in all other joints in the kinematic chain. As we can see from Figure 4.7, the ROMP technique suffers from great instability in the root joint (the value of jitter equals 0.5 radians). Such problems with instability in the root joint are responsible for global movement instability in early experiments; however, due to temporal and SLAM world grounding, WHAM reaches near-zero root joint jitter level (0.02 radians).

Therefore, WHAM allows creating a ground truth for joint angle estimation.

Chapter 5

Timeline and Resource Required

5.1 Timeline

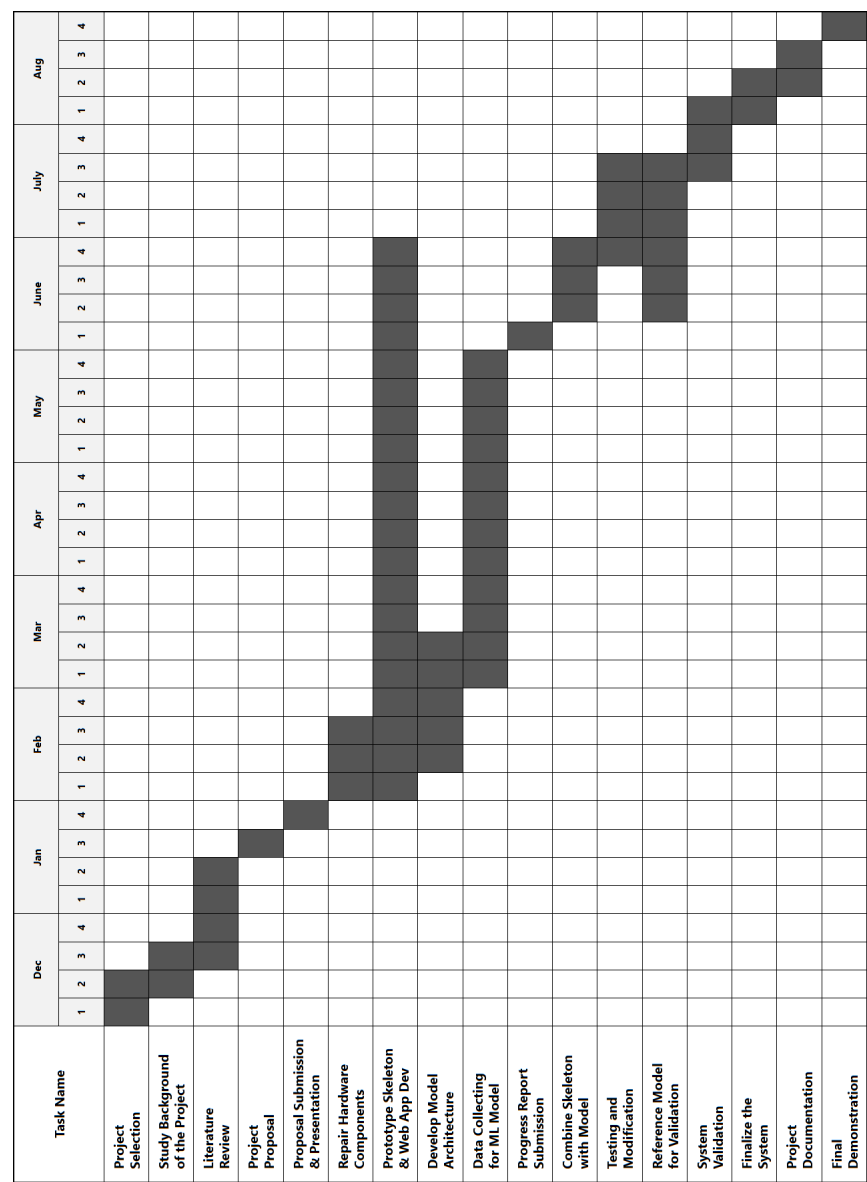


Figure 5.1: TimeLine

5.2 Resource Required

This is a general estimation of budget for this project. These prices can be vary since the exact components and services are not decided yet.

Table 5.1: Estimated Budget for Project Components and Services

No	Item / Service	price
1	Repair for existing components	5 000
2	Web hosting	15 000
3	ML training platform	8 000
4	Cloud computing	20 000
5	API s	10 000
	Total	58 000

Chapter 6

Conclusion

This project proposal offers a framework for accomplishing accurate human activity recognition using a sparse sensor configuration of five 3-axis accelerometers. Through the placement of the sensors at the wrist, ankle, and torso, the proposed approach meets the requirement for developing a non-invasive and portable solution for human activity recognition, which requires lower costs compared to the usual motion capture systems that are dense, expensive, and complex.

The approach systematically investigates three different estimation techniques in parallel to connect raw inertial measurements with realistic visualization. They are pattern recognition through machine-learning techniques, orientation equations in geometry, and inverse kinematics theories. The combination of different techniques can improve the reliability of joint posture and orientation estimation in the presence of ambiguities in raw accelerometer data.

One of the key contributions of the above-mentioned research is the creation of a personalized “Digital Twin” using an “automatic three-dimensional avatar generation pipeline.” The process eliminates the need for generic digital templates and manual measurements in place and uses deep regression techniques like ROMP or HMR in conjunction with the Skinned Multi-Person Linear (SMPL) model, thereby achieving metric accuracy and realistic skin deformation while avoiding joint volume loss and kinematic drift problems.

Finally, the solution also applies an effective four-tier validation mechanism that takes into account the geometric, physical, perceptual, and reliability verification to ensure the performance of the system aligns well with the expectations in the real world. This project’s effective integration of sparse inertial sensing with advanced computer vision and biomechanical modeling, results in a scalable and efficient solution for applications in healthcare, sports analysis, and immersive virtual environments.

References

- [1] X. Yi, Y. Zhou, and F. Xu, “Transpose: Real-time 3d human translation and pose estimation with six inertial sensors,” *ACM Transactions On Graphics (TOG)*, vol. 40, no. 4, pp. 1–13, 2021.
- [2] X. Xiao, J. Wang, P. Feng, A. Gong, X. Zhang, and J. Zhang, “Fast human motion reconstruction from sparse inertial measurement units considering the human shape,” *Nature Communications*, vol. 15, no. 1, p. 2423, 2024.
- [3] Y. Huang, M. Kaufmann, E. Aksan, M. J. Black, O. Hilliges, and G. Pons-Moll, “Deep inertial poser: Learning to reconstruct human pose from sparse inertial measurements in real time,” *ACM Transactions on Graphics (TOG)*, vol. 37, no. 6, pp. 1–15, 2018.
- [4] J. Tautges, A. Zinke, B. Krüger, J. Baumann, A. Weber, T. Helten, M. Müller, H.-P. Seidel, and B. Eberhardt, “Motion reconstruction using sparse accelerometer data,” *ACM Transactions on Graphics (ToG)*, vol. 30, no. 3, pp. 1–12, 2011.
- [5] P. Kelly, C. Ó Conaire, and N. E. O’Connor, “Human motion reconstruction using wearable accelerometers,” 2010.
- [6] S. Gunawardena, P. Amarasingha, S. Gunawardhana, and S. Sabaragamuwa, “Wireless sensor network aided human activity detection using artificial intelligence,” June 2025. Preprint on ResearchGate.
- [7] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, “SMPL: A skinned multi-person linear model,” *ACM Transactions on Graphics (TOG)*, vol. 34, no. 6, pp. 1–16, 2015.
- [8] A. Kanazawa, M. J. Black, D. W. Jacobs, and E. Enqvist, “End-to-end recovery of human shape and pose,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7122–7131, 2018.
- [9] Y. Sun, Q. Bao, W. Liu, W. Gao, Y. Fu, M. Chen, and T. Mei, “Monocular, one-stage, regression of multiple 3d people,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 11179–11188, 2021.
- [10] S. Jung, J. Sun, M. Kim, J.-S. Park, J. Park, and K. M. Lee, “Wham: Reconstructing world-grounded humans with pose and shape from video,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1–15, 2024.

- [11] S. Saito, Z. Huang, R. Natsume, S. Morishima, A. Kanazawa, and H. Li, “Pifu: Pixel-aligned implicit function for high-resolution clothed human digitization,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 2304–2314, 2019.
- [12] J. P. Lewis, M. Cordner, and N. Fong, “Pose space deformation: a unified approach to shape interpolation and skeleton-driven deformation,” in *Proceedings of the 27th annual conference on Computer graphics and interactive techniques (SIGGRAPH)*, pp. 165–172, 2000.
- [13] S. Zhao, Z. Wang, T. Luan, J. Jia, W. Zhu, J. Luo, J. Yuan, and N. Xi, “Pp-motion: Physical-perceptual fidelity evaluation for human motion generation,” *arXiv preprint arXiv:2508.08179*, 2025.
- [14] S. Zhu, Y. Li, J. Li, Q. Wu, Z. Wang, H. Ma, and W. Liang, “Human motion estimation with everyday wearables,” *arXiv preprint arXiv:2512.21209*, 2025.
- [15] J. L. Ponton, E. Alvarado, L. G. Foo, N. Pelechano, C. Andujar, and M. Habermann, “Step2motion: Locomotion reconstruction from pressure sensing insoles,” *arXiv preprint arXiv:2510.22712*, 2025.
- [16] P. Molloy, Y. Zhou, M. Habermann, and C. Theobalt, “The impact of anthropometric scaling on inertial motion capture accuracy,” *Journal of Biomechanics*, vol. 152, p. 111571, 2023.
- [17] B. D. Lucas and T. Kanade, “An iterative image registration technique with an application to stereo vision,” in *Proceedings of the 7th International Joint Conference on Artificial Intelligence (IJCAI)*, vol. 2, pp. 674–679, 1981.
- [18] K. Zuiderveld, “Contrast limited adaptive histogram equalization,” in *Graphics Gems IV*, pp. 474–485, Academic Press Professional, Inc., 1994.
- [19] K. He, J. Sun, and X. Tang, “Guided image filtering,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 35, no. 6, pp. 1397–1409, 2013.
- [20] Y. Xu, J. Zhang, Q. Zhang, and D. Tao, “ViTPose: Simple vision transformer baselines for human pose estimation,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, pp. 38571–38584, 2022.
- [21] X. Wang, L. Xie, C. Dong, and Y. Shan, “Real-ESRGAN: Training real-world blind super-resolution with pure synthetic data,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 1905–1914, 2021.
- [22] S. Goel, G. Pavlakos, J. Rajasegaran, A. Kanazawa, and J. Malik, “Humans in 4D: Reconstructing and tracking humans with transformers,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 14983–14994, 2023.
- [23] Y. Li, H. Mao, R. Girshick, and K. He, “Exploring plain vision transformer backbones for object detection,” in *European Conference on Computer Vision (ECCV)*, pp. 280–296, Springer, 2022.